

SeeSawBot: An LLM-Driven Chatbot Mediating Across Private and Shared Slack Channels to Support Team Dynamics

Yihe Wang

ywan1125@ucsc.edu

University of California Santa Cruz
Santa Cruz, California, USA

Kehua Lei

klei4@ucsc.edu

University of California Santa Cruz
Santa Cruz, California, USA

Sheng-Yang Chiu

sheng-yang_chiu@sfu.ca

Simon Fraser University
Burnaby, British Columbia, Canada

Katherine Isbister

kisbiste@ucsc.edu

University of California Santa Cruz
Santa Cruz, California, USA

David Lee

dlee105@ucsc.edu

University of California Santa Cruz
Santa Cruz, California, USA

Kathryn E. Ringland

kringlan@ucsc.edu

University of California Santa Cruz
Santa Cruz, California, USA

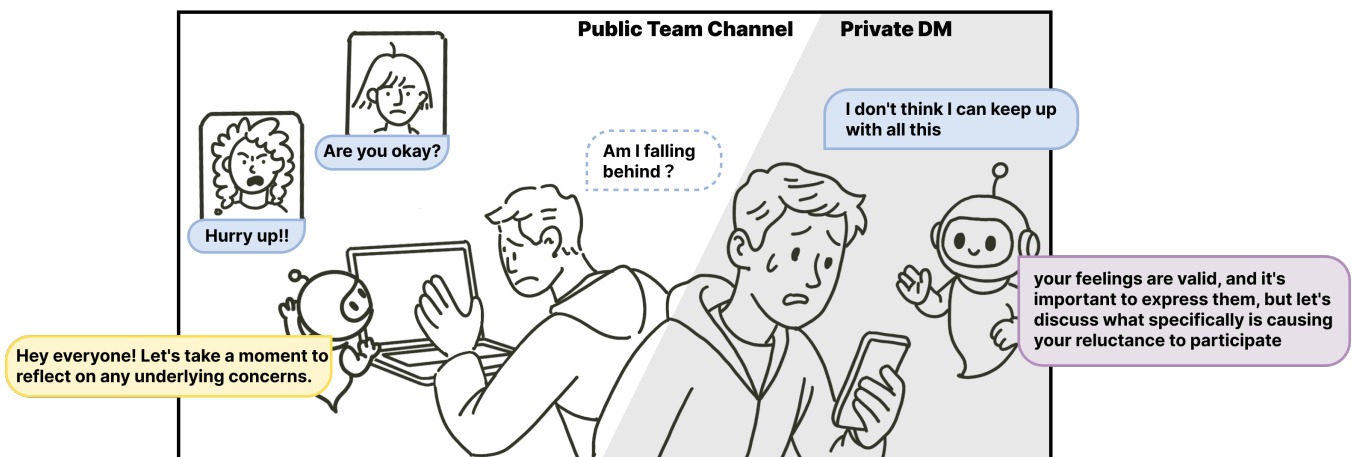


Figure 1: SeeSawBot mediates across private DMs and public team channels, enabling cross-space interventions that support team dynamics and collaboration.

Abstract

While conversational agents increasingly mediate teamwork, prior work has mainly focused on *when*, *what*, or *to whom* an intervention is directed, with little attention to *where* mediation occurs. Therefore, we introduce SeeSawBot, an LLM-driven chatbot that operates across private DMs and public channels. Following a formative study, we deployed SeeSawBot in student Slack teams as a *technology probe* for eight weeks, collecting bi-weekly reflection surveys and post-deployment interviews. Findings show that cross-space mediation fostered sense-making across private and public spaces and redistributed emotional labor through interventions that played different relational roles over team development. We discuss cross-space mediation as both a *boundary object* and *boundary actor*, and argue that future evaluation frameworks should capture *relational agency* by attending to the back-and-forth negotiations through which groups construct collective understanding.

We conclude with design implications that foreground *where* as a variable for future computational mediators, a seesaw of agency and autonomy.

CCS Concepts

• **Human-centered computing** → HCI theory, concepts and models; Empirical studies in collaborative and social computing.

Keywords

Conversational Agent, Multi-Party Conversation, Proactive AI, Group Processes, Research through Design, Technology Probe

ACM Reference Format:

Yihe Wang, Kehua Lei, Sheng-Yang Chiu, Katherine Isbister, David Lee, and Kathryn E. Ringland. 2026. SeeSawBot: An LLM-Driven Chatbot Mediating Across Private and Shared Slack Channels to Support Team Dynamics. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 23 pages. <https://doi.org/10.1145/3772318.3791880>



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2278-3/2026/04

<https://doi.org/10.1145/3772318.3791880>

1 Introduction

People communicate differently depending on the situation. For example, concerns might be shared quietly in private before being

raised in a group, or disagreements may be worked through one-on-one before being discussed publicly. These shifts affect how people open up, coordinate with others, and repair relationships. Sociologists have long described this dynamic. Goffman's notion of frontstage and backstage [34] captures how people modulate self-presentation in public versus private arenas, while Simmel's work on secrecy highlights selective disclosure as a constitutive mechanism of social life [97]. Together, these theories suggest that communication depends not only on timing and content, but also on spatial placement.

Despite growing interest in computational mediation, HCI research has not fully engaged with the spatial dimensions of communication. Prior work has explored structuring group flow (e.g., agenda scaffolding [59], decision prompts [39], turn-taking support [52]), personalizing interactions (e.g., using social media for introductions [94], tailoring prompts to user context [32, 109]), and timing interventions (e.g., detecting voice tension [14]). However, these systems tend to focus on the familiar “3W” framework [67] (*when*, *what*, and *whom* to intervene). These dimensions overlook the fundamental question: *where* should intervention occur? Most mediators are built either for public spaces (multi-user exchanges; e.g., [59, 62]) or private spaces (one-on-one chats; e.g., [14, 78]), missing moments where mediation must traverse spaces to negotiate relational dynamics.

Therefore, we developed *SeeSawBot*, an LLM-driven Slack chatbot that navigates conversations across multiple conversational contexts, especially private DMs (direct messages) and public channels, as illustrated in **Figure 1**. We deployed it as a *technology probe* [43] to reveal how *group processes* [115] unfold with cross-spaces computational mediation. By using it as a *technology probe*, we approach this as both a technical and cultural question: **How do group processes unfold when an AI mediator engages with participants across different conversational spaces, such as private DMs and public channels?**

This study, based on undergraduate classroom teams, used a two-phase design to examine how cross-space mediation reshapes group dynamics and perceptions of AI. First, we conducted a formative mock-up study ($n=10$) to inform the design of *SeeSawBot*. We then deployed *SeeSawBot* in student Slack teams for eight weeks. Our analysis drew on bi-weekly reflection surveys ($n=63$, 51, 57, 52) and post-deployment interviews ($n=9$). Our study shows that cross-space mediation can transform team dynamics in three ways. First, it enables new information flows by bridging private reflection and public discussion, as well as tailoring group activity into individualized guidance. Second, it redistributes emotional labor through roles such as gateway for hesitant voices, a balancer of participation, a social buffer absorbing socially risky tasks, and a reflective mirror supporting awareness and closure. Third, it reveals that the spatial placement of interventions shapes perceptions of autonomy, agency, and legitimacy. These findings advance understanding of human-AI teamwork by foregrounding *where* mediation occurs as a critical design variable, positioning computational agents not only as *boundary objects* that create shared artifacts but also as *boundary actors* that actively reconfigure participation and authority.

The rest of this paper unfolds as follows. In **Section 2**, we review work on group processes and AI-mediated communication to highlight a key gap: the lack of systems that navigate both private

and public interaction spaces. We then introduce the technology probe as our guiding methodological approach (**Section 3**), followed by our formative mock-up test and its design implications for our technology probe (**Section 4**). Building on these insights, we present the design of *SeeSawBot* (**Section 5**). **Section 6** describes our 8-week deployment study, including both survey reflections and post-deployment interviews. We then present findings from these sources in **Section 7**. In **Section 8**, we discuss the broader implications of cross-space AI mediation, and conclude by acknowledging our limitations and future directions in **Section 10**.

2 Background

2.1 Social Mediation in Group Processes

Teams are not static collections of individuals but dynamic social systems whose members must continually negotiate how to work together [46]. Classic theories of group development, such as Tuckman's five-stage model (forming, storming, norming, performing, and adjourning), illustrate how coordination, conflict, cohesion, identity, and leadership emerge and shift over time [102, 103]. These phenomena are often discussed under the umbrella of *Group processes* [115], highlighting how interactional practices transform individual contributions into collective outcomes.

Mediation refers to a process where parties work with a neutral third party who helps them negotiate their differences [106]. In practice, mediation is often associated with conflict resolution, where a facilitator restores strained relationships by guiding exchanges and enabling negotiation [30, 68]. However, social mediation also encompasses broader efforts to foster or rebuild group cohesion [47, 104]. Here, mediators may be formally appointed (e.g., instructors, facilitators, designated leaders) or emerge informally (e.g., respected peers, senior members, or rotating “process stewards”) [72]. Their roles extend beyond managing conflict [47]. They also redistribute workloads, clarify expectations, maintain a constructive group climate [104], and contain reputational risks or emotional tensions that might derail negotiations [29].

To do this, mediators draw on practices that range from structuring the overall group process to intervening in the moment-to-moment conversation. Structurally, they shape group processes by setting agendas, defining roles, enforcing norms, and facilitating reflection [104, 106]. In real time, they use conversational strategies such as facilitative moves [3], reframing disagreements to reduce defensiveness [72], or moving sensitive topics into private backchannels to preserve trust [18, 73, 89]. These practices can be grouped into three broad categories [56]: reflexive interventions, which establish the groundwork. Contextual interventions, which foster a climate for constructive dialogue and problem solving. And substantive interventions, which address the issues in dispute directly.

As collaboration increasingly shifts to digital environments [19, 66], teams face new challenges, such as managing trust without physical presence, mitigating misunderstandings across asynchronous channels, and fostering cohesion despite reduced social cues [19, 83]. At the same time, online environments offer new opportunities. Digital platforms enable mediators to document processes, facilitate asynchronous reflection, and use features like breakout rooms or chat to manage sensitive exchanges [19, 66]. In this context, mediation practices must be reimaged for virtual settings to sustain trust and learning in distributed teams [66, 83].

2.2 Computational Mediators for Group Processes

In the last decade, researchers have begun to explore *computational mediators* that explicitly facilitate *group processes*, typically instantiated as conversational agents [53, 78, 86, 88] or social robots [35, 93]. Early work in human-robot interaction examined social robots as embodied mediators in face-to-face settings. For example, Shen et al. [93] demonstrated that a robot could intervene in preschoolers' disputes by providing prompts that reduced verbal aggression and encouraged resolution. More recent systems such as ROMEO [35] integrate verbal and non-verbal cues, including haptic feedback, to manage emotional climates and facilitate inclusive participation in group tasks.

Building on this line of inquiry, HCI researchers have investigated chatbots as lightweight, text-based mediators in online or distributed collaboration. One line of work emphasizes structuring interaction, for example, by organizing proposals in collaborative modeling [86] or balancing participation [53]. Mediation can also mean negotiating social norms, as in MediationBot, which facilitated agreement in multiparty privacy conflicts [88]. Others support social connection [45, 94, 95], such as BlahBlahBot, which improved conversational quality between strangers by suggesting personalized topics [95], or emotional awareness in distributed teams [6]. Wang et al. highlight online learners' demand for targeted private nudges into social encounters [109]. Chow et al. further show that openness to computational mediation is relational and develops over time, is highly space sensitive with intimate content better suited to semi private channels, and is shaped not only by interpersonal closeness but also by perceived power risks [16]. A closely related direction examines AI mediators in synchronous meetings. Houtti et al. [41] introduced a virtual co-host that privately messaged over-participants and hosts during video meetings, and found that, although its feedback often "fell on deaf ears", such private, gentle interventions were experienced as socially comfortable.

Across both embodied and text-based modalities, interventions typically rely on carefully engineered context signals, such as dialogue structure [53, 93], emotional cues [6, 35], or social preferences [88] to guide when and how to intervene. Because these cues are encoded as rules or domain-specific features, mediators often struggle to adapt when situations deviate from their original design. Moreover, most systems have targeted narrow aspects of interaction (e.g., turn-taking, conflict prompts, emotional regulation) and have been evaluated in small-scale or short-term deployments.

2.3 Human-AI Teamwork

Recent advances in large language models have shifted computational agents from scripted assistants to more adaptive teammates capable of participating in group work. Rather than framing agency as a zero-sum tradeoff where automation reduces human control, recent research emphasizes augmentation. AI can support human judgment like a junior colleague [28, 64]. Shneiderman's Human-Centered AI framework models human control and computer autonomy as independent axes [107]. Building on this, Kim et al. [54] classified AI roles as tools, servants, assistants, and mediators. Mediators combine high autonomy with human involvement and have been shown to foster greater credibility and acceptance [54].

Human-AI teamwork offers value through complementary strengths [116]. Humans contribute contextual judgment, empathy, and norm-setting, while AI can surface patterns, sustain attention, and redistribute interactional labor [37, 110]. As a result, researchers have begun designing AI partners that support not just task efficiency but also the deeper social and cognitive dynamics of collaboration, such as aligning intentions and sustaining mutual engagement [37, 77]. Group collaboration, however, unlike individual productivity, unfolds in dynamic, multi-party settings where agency must be continuously negotiated [31].

A central design tension is whether AI acts *proactively* (initiating actions without being asked) or *reactively* (waiting for human inputs). Reactive systems (e.g., [89]) preserve user autonomy but burden humans with detection and response. Proactive systems (e.g., [12, 14, 52, 77]) ease that burden and help sustain momentum, but they risk mistimed interventions or perceived overreach [77]. The trade-off, as Oh et al. [77] noted, lies between efficiency (proactive reduces users' cognitive load by taking initiative) and autonomy (reactive systems or those that seek permission before acting preserve human agency). Therefore, as an intelligent mediator, it must negotiate initiative, autonomy, and collaboration within multi-party settings rather than act as a static assistant. This perspective directly informed our proactive and reactive design choices (Figure 4) and the baseline and probe configurations described in Section 5.1.

2.4 Spatial Dimension of Communication

Where communication happens is not neutral. It shapes meaning and social relations. Goffman's dramaturgical framework highlights how *frontstage* behavior occurs in the presence of an audience, where individuals align with expected roles and maintain an idealized image, whereas *backstage* spaces allow for candid exchanges and relational repair among teammates [34]. Mediators routinely navigate these spaces through techniques such as caucusing, holding private conversations with each party before or during group dialogue to balance trust with collective clarity [17]. In digital settings, Harrison and Dourish [38] argue that interaction does not just occur in space but they construct place, as environments become meaningful through shared norms and expectations. This movement is consequential. Theories of privacy boundaries [82] and contextual integrity [75] emphasize that people actively manage how information flows across spaces, deciding what stays private, what goes public, and when to shift those boundaries.

Backchannel communication refers to informal, often discreet exchanges that run parallel to the main group discourse. Across classrooms, lectures, and conferences, research consistently shows that *backchannels* broaden participation, provide affective support, and enable subtle coordination, but also raise tensions around distraction, equity, and visibility [25, 84, 87, 101]. For example, Class-Commons demonstrated how a classroom backchannel allowed students to validate one another and ask questions that might never surface in the *frontchannel*, thereby increasing inclusivity while also shifting classroom dynamics [25]. Designers have begun to treat these channels as configurable spatial strata. Some, like Whisper-Channel, explored low-volume audio cues to let remote participants "whisper" into a live meeting, fostering connection but also surfacing new concerns about privacy [73]. Others, like FluidMeet, allowed fluid movement between in-group, between-group, and private conversations, reducing isolation while making boundaries

more permeable [42]. Such work highlights that spatial placement can alter whether a message functions as a supportive aside, a reputational call-out, or a scaffold for group awareness. This spatialization of communication extends beyond short lived encounters. In persistent communities such as Discord, different roles see different channels. Admin channels concentrate decision making and coordination, while topic channels host everyday interaction, distributing power and attention [90]. On Twitch, chat bots flag spam or rule violations, but their triggers and actions are set by admins, making them semi-autonomous, but their thresholds and actions are configured by human moderators, positioning them as semi autonomous systems that reinforce norms [91]. Metaverse workspaces push this logic to the architectural level, where room placement, environmental visibility, and avatar presence shape how teams negotiate co-presence, access, and trust [80, 81]. In designing SeeSawBot, our aim was not only to build a functional mediator, but also to examine how spatial dynamics shape mediated collaboration, treating space not as a neutral backdrop, but as a site of ongoing social negotiation.

3 Method Overview

3.1 Technology Probe as Method

We adopted a *technology probe* approach [43] to study how people appropriate emerging technologies in everyday group work. Probes are deliberately partial and open-ended. Their purpose is not to test a finished product, but to provoke exploration, invite adaptation, and surface unanticipated practices [15, 40, 48]. Therefore, our SeeSawBot was deployed as an evolving probe rather than a completed system, iterating SeeSawBot throughout this study, incorporating participant feedback and responding to emergent use. This approach aligns with the exploratory nature of this study, which we seek to introduce *where* mediation occurs (DMs vs. channels) as a new design variable. This iterative stance is central to the value of technology probes: to expose possibilities, prompt reflection, and co-shape futures through use in progress [13, 99].

Because, to our best knowledge, no established benchmarks exist for assessing this dimension, we foregrounded qualitative data to understand how participants interpreted and evaluated interventions. Drawing from surveys, interviews, and researcher memos, we examined how cross-space mediation influenced group processes and dynamics. These situated accounts not only illuminate how SeeSawBot functioned in practice but also provide the foundation for a future evaluation framework tailored to space-sensitive human-AI collaboration.

3.2 Study Context

We focus on college student teams in classroom settings, a common choice in HCI and CSCW because they offer accessible, bounded, and semi-naturalistic contexts for exploratory deployments [33]. Plus, the classroom enabled a term-long “living lab” deployment, where teams pursued authentic goals while researchers could observe under controlled conditions [43]. In student teams, members are in a stage of emerging adulthood with digital fluency but uneven maturity and teamwork experience [2, 76]. Unlike professional teams, classroom groups are normally short-lived, externally assigned, and deadline-driven [76], so coordination challenges often arise before psychological safety and trust are fully

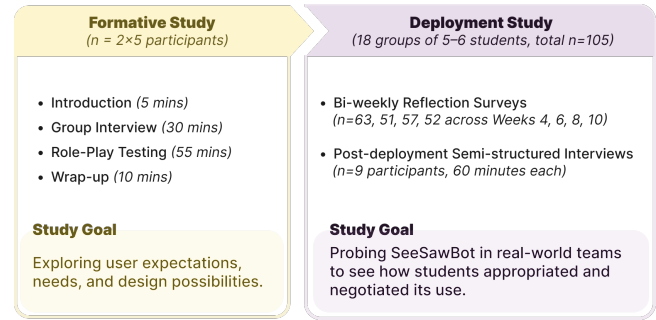


Figure 2: The project used a two phase empirical design. Phase I (Formative Study) employed focus group role play to surface breakdowns, expectations, and boundaries in cross-space conversations. Phase II (Deployment Study) introduced SeeSawBot into 18 undergraduate Slack teams for eight weeks and collected longitudinal feedback through bi-weekly reflection surveys and post deployment interviews.

established [26], workload imbalances and fairness concerns are common [76], and grades and peer recognition are structured as central motivators [105]. Despite these differences, student teams exhibit authentic teamwork dynamics such as conflict, leadership, and coordination breakdowns that make them suitable for probing novel interventions [33].

3.3 Study Phases

The study consisted of three sequential phases (Figure 2):

- **Formative Study** (see Section 4): Conducted focus group discussions with experienced teaching assistants and students to surface expectations, prior experiences with conflict, and initial impressions of AI-mediated support.
- **Deployment Study** (see Section 6): Combined two complementary qualitative approaches: *bi-weekly reflection surveys* and *post-deployment interviews*.
 - **Bi-weekly reflection surveys** (see Section 6.2): During the 8-week deployment of SeeSawBot in student team Slack channels, participants submitted bi-weekly reflections on communication dynamics and AI interactions, producing in-situ and longitudinal accounts across a wider participant set.
 - **Post-deployment interviews** (see Section 6.3): Following the deployment, we conducted semi-structured interviews to elicit retrospective, interpretive accounts, with particular attention to emotional, ethical, and contextual responses to SeeSawBot’s presence and interventions. While both produced qualitative textual data, the surveys emphasized breadth and temporal snapshots, whereas the interviews provided depth and interpretive nuance. Together, they offered a complementary view of how SeeSawBot was appropriated over time.

All study procedures were reviewed and approved by the university’s Institutional Review Board (IRB).

G#	P#	Gender	Age	Year in School	Experience
G1	FP1	Female	27	Graduate	Mediator
	FP2	Female	20	Junior	Member
G2	FP3	Male	29	Graduate	Both
	FP4	Female	34	Graduate	Member
G3	FP5	Male	24	Graduate	Member
	FP6	Female	21	Junior	Both
G4	FP7	Female	45	Graduate	Mediator
	FP8	Male	28	Graduate	Both
G5	FP9	Female	29	Graduate	Mediator
	FP10	Male	24	Graduate	Both

Table 1: Demographic information for formative study participants.

4 Phase I: Formative Study

Before full deployment, we ran a formative mock-up test of SeeSawBot in a controlled, simulated team conversation on Slack, to explore how it might operate across DMs and public channels. The simulation helped us examine key challenges such as how teams coordinate across channels, how they respond to tension, and how a third-party AI might intervene constructively. These insights guided early design refinements, allowing us to shape a prototype that aligned with user expectations while also surfacing rich, emergent social behaviors. At this phase, SeeSawBot was still under active development and undergoing significant iteration. Nonetheless, the prototype demonstrated three foundational capabilities:

- It functioned across both private DMs and public channels.
- It picked up on conversational cues to provide timely social support.
- It remained in the background when not needed.

We guided our analysis with the research question: How do participants use and imagine SeeSawBot across private and public spaces, and what can their expectations, challenges, and ideas reveal for technology probe design?

4.1 Participants

We recruited participants from the authors’ university in the United States with prior teamwork experience, especially those who had engaged with difficult *group processes*, including coordination challenges, disagreements, and conflict resolution. Five focus groups (10 participants: 6 female, 4 male; aged 20–45) participated in the mock-up testing (Table 1). Seven had served as mediators in past team settings, and seven had been team members in difficult conversations requiring mediation. This mix ensured that discussions were grounded in both the strategic perspective of facilitators and the lived experience of team members.

4.2 Procedure

We created a dedicated Slack workspace for all mock-up testing sessions, with a separate channel for each group. Participants joined using their own Slack accounts and kept Slack open throughout

the session. All sessions were held remotely via Zoom, with two researchers (host and moderator) facilitating. Each 90-minute session followed this sequence:

- (1) **Introduction (5 minutes)** walked participants through the study goals, session flow, and confidentiality guidelines.
- (2) **Group Interview (30 minutes)** explored participants’ past teamwork experiences, focusing on group processes, coordination, and handling challenging conversations, including any third-party involvement.
- (3) **Role-Play Testing (55 minutes)** engaged participants in four scripted Slack scenarios representing different stages of collaboration (e.g., team formation, deadline pressure, disagreement, final delivery), interacting with SeeSawBot in both public and private channels, then reflecting on its interventions in terms of the timing, tone, and cross-space behavior.
- (4) **Wrap-up (10 minutes)** concluded with open discussion on impressions, potential applications, and expectations for AI-mediated teamwork.

We collected observational data and researcher memos. All sessions were audio-recorded and transcribed.

4.3 Analysis and Findings

We conducted a thematic analysis (see Appendix B) of researcher memos and focus group transcripts. Our analysis focused on group dynamics, especially moments when participants expressed divergent expectations, navigated boundary tensions, or developed new practices. Following Braun and Clarke’s view that variation reveals meaningful patterns [8, 9], and Khovanskaya et al.’s framing of frictions as design signals [51], we treated these divergences not as critical insights for system design. Below, we summarize participant perspectives in each thematic area and present related design principles (DPs) for our technology probe.

4.3.1 Task-Focused vs. Emotionally Supportive (40 mentions). Participants were divided in what they wanted from SeeSawBot. Some preferred a professional, task-oriented style focused on “actionable goals”, while others wanted a more emotional calibration, such as support in self-regulation (e.g., help with “word choices” or “control myself not being rude”), feeling understood (“someone understands me”), and comfort (“teddy bear”). These contrasting needs suggest that SeeSawBot’s effectiveness depends less on researchers pre-determining the “right” persona and more on its ability to flexibly adapt to diverse human needs and shifting group dynamics.

DP-1 The probe should support fluid transitions between efficiency and reflection through conversational interaction, enabling the AI to sense user needs contextually and adjust its response style accordingly.

4.3.2 Boundary-Testing Provocations (46 mentions). Participants reacted in varied ways to how SeeSawBot navigated between public channels and private messages. For some, clear boundaries were essential, especially the expectation that private DMs should remain confidential: (“I thought it should be private conversation between me and the test bot”) Others welcomed the AI referencing private exchanges in public, particularly when it offered visible support or

validated their perspective: (“like having a second person to root for me (after DM communication)[...] if I can make the chatbot root for me, I will definitely invite it into the chat.”) Still, the most appreciated behavior was the reverse, especially when the AI noticed something in a public thread and followed up privately. “When there’s no one wants to work, I have to do it myself. It says, ‘feel like you’re feeling the weight of the project on your shoulders.’ [...] I feel like someone is understanding me.” These interventions felt more respectful, less exposing, and more emotionally attuned.

DP-2 The probe can use public content in DMs. DM content may also be used, but only with great care. It should never be shared publicly or with others unless explicitly invited, and even private use must align with user expectations and context.

4.3.3 Authority Positioning Flexibility (32 mentions). Participants had differing expectations for how authoritative the AI should be. Some preferred a quiet, facilitative assistant. One that offered reminders and relayed others’ input without making decisions. Others wanted a more assertive AI to enforce accountability and speak up when human members hesitated: “I like that the chatbot is kind of like being the bad guy here, telling people what’s doing wrong.” Rather than reflecting fixed preferences, participants described authority needs as situational. Many welcomed gentle support for routine tasks: “just a gentle reminder, other people are worrying about this” or “automatically set the calendar invite.” In contrast, others wanted the AI to step in when groups stalled or avoided conflict: “calling out the people that weren’t responding more” or “give pressures for that person [who hasn’t spoken in 5 minutes].” Notably, some saw assertiveness not as dominance, but as helpful delegation: “having a bot to do it than having either one of us to do it sounds a better plan.”

DP-3 The probe should dynamically adjust authority levels, testing when assertive intervention is welcomed versus when facilitative support is preferred.

4.3.4 Proactive or Reactive Intervention Strategies (35 mentions). Participants imagined multiple ways SeeSawBot could support group dynamics, ranging from autonomous monitoring to user-directed prompts. These scenarios reflected flexible roles for the AI, depending on how group conversations unfolded. Some participants described the AI taking initiative during moments of silence or tension. Participants described scenarios where the AI would monitor the conversation and step in when activity dropped or tensions rose. They imagined it “send[ing] a message to whoever is not responding” or gently pressuring members who had gone silent. In more sensitive situations, the AI could surface frustrations others were reluctant to voice: “maybe the bot can say, ‘Hey, you’re not very cooperative.’” Some viewed this as a way to avoid being “the bad guy” themselves. Others envisioned a more user-driven model, where the AI acts when prompted. In these scenarios, participants wanted to DM the AI with instructions like “help me get everybody working” or “point out someone’s not participating.” Some emphasized that they saw the AI as a commandable tool that they could

activate to handle group issues discreetly and efficiently. One participant proposed a hybrid strategy: the AI could check in privately before taking public action: “If there’s no one responding, maybe the agent DMs me and helps me articulate my frustration, and then gives me some options.” These imagined roles suggest participants valued flexibility, imagining both autonomous and user-directed AI behavior based on group dynamics.

DP-4 The probe should test both proactive check-ins and reactive responses to capture different user needs and optimal intervention timing.

5 Design of SeeSawBot: The Technology Probe

Our formative study showed that teamwork requires flexibility beyond any fixed mediation style, so we derived four principles framing SeeSawBot as a probe that adapts with context.

5.1 Probe Configurations and Reporting

We implemented SeeSawBot as a chatbot in a Slack workspace with multiple team-specific channels. We implemented SeeSawBot using the GPT-4o model and deployed the system on Google Cloud for the duration of the study. Users could tag SeeSawBot in a channel for reactive support, message it directly for private replies, or let it listen and step in proactively in channels or DMs. **Figure 4** illustrates these two interaction modes, highlighting the bot’s balance of proactive and reactive responses [\[DP-4\]](#).

As a technology probe, SeeSawBot was not fixed at launch but evolved throughout deployment. We began with a minimal baseline, V1. The chatbot monitored group and private messages, analyzing recent content to detect signs of confusion, hesitation, or disagreement. When triggered, it issued short, context-specific interventions, typically clarifying questions or suggestions to expand on prior statements. These were delivered either publicly in channels or privately in DMs or both, based on the emotional tone and relevance of the topic. For example, when a group thread showed signs of confusion, the bot followed up privately to prompt reflection (see **Figure 7**). V1 also introduced two scheduled behaviors: 1) Weekly to-do reminders posted in team channels—a preset list of assignments sent at a predetermined time; and 2) Weekly recap messages delivered via DM—a summary of accomplishments and an interpretation of group dynamics based on the past week’s conversations, also sent at a scheduled time. These two features were incorporated into the chatbot to encourage students to engage with it in private interactions. In V2, the bot began tracking participation, detecting quiet members who had relevant context and had not been recently active or nudged before. It then nudged them privately or in-channel depending on context. In V3, the bot recognized assignment-related discussions and responded with information aligned to course specifications. In V4, it used historical patterns of interaction to suggest follow-ups between specific members, helping teams re-engage inactive dyads or resolve persistent gaps in coordination. **Table 2** summarizes the concrete iterations that participants encountered.

Due to the non-deterministic execution of LLM-based systems, SeeSawBot exhibited occasional bugs, unintended behaviors, and

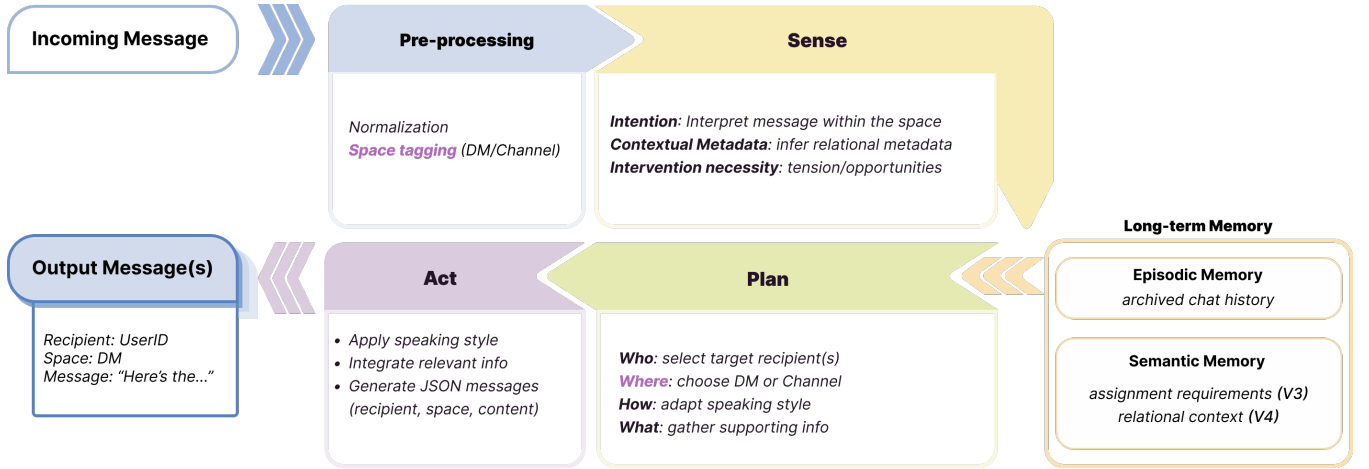


Figure 3: Design of SeeSawBot’s sense-plan-act pipeline. The system first pre-processes incoming messages through normalization and space tagging, then interprets them using contextual and relational cues during the sense stage. Guided by episodic and semantic memory, the plan stage decides whether to intervene and selects an appropriate target and space. The act stage generates tailored responses and dispatches them across DM and channel contexts.

execution inconsistencies (e.g., when participants were coordinating in person but used channels primarily to share documents or assignment artifacts rather than conversational exchange, SeeSawBot sometimes misinterpreted it as a signal for intervention). We do not exhaustively document all such issues. Instead, we focus on reporting interventions that shaped how participants interpreted and engaged with SeeSawBot’s mediating role.

Version	Launch Week	Description of changes
V1	Week 2	- Baseline - Weekly DM chat recap. - Weekly Channel to-do reminder
V2	Week 4	Participation balancing.
V3	Week 6	Assignment-aware answers begin.
V4	Week 8	Relationship-aware reasoning.

Table 2: Iteration history of SeeSawBot probe over the 8-week deployment. Each version (V1-V4) progressively layered new functions on top of the baseline (weekly DM recap + weekly channel reminder), illustrating how the system evolved in response to survey feedback and emergent use.

5.2 SeeSawBot Pipeline and Modules

SeeSawBot followed a *sense-plan-act* pipeline (see **Figure 3**), inspired by classical robotics architectures [10, 74] and conceptually related to recent LLM-based agentic systems that employ sequential *perceive-plan-act* structures [4, 21, 61, 62]. This pipeline provided a consistent backbone across the deployment, while specific modules were progressively extended through probe iterations (**Table 2**).

5.2.1 Preprocessing and Triggers. Each incoming Slack message was first parsed into a structured representation with metadata

(sender, timestamp, space label). This representation served as the structured input for subsequent pipeline stages.

5.2.2 Sense: Context Parsing. In the sensing stage, the preprocessed message and its metadata (e.g., space label, sender, timestamp) were analyzed for both task-related and affective cues. To keep this stage lightweight, it operated only over a short sliding context window, drawing on the most recent exchanges rather than the full history. For channel messages, this stage first performed an INTERVENTION_DECISION: assessing whether the situation warranted SeeSawBot’s involvement in the ongoing conversation. When intervention was considered [DP-4], the system then performed a MESSAGE_RESTATEMENT, extracting key content and user intentions to inform later planning. Because sensing was invoked frequently, this module was deliberately kept lightweight, reserving LLM inference for cases that required additional nuance [DP-1].

From V2 onward, sensing was extended with *participation balancing*, and was further enriched in V4 with *relationship-aware reasoning*. Participation balancing monitored relative engagement levels across members. When persistent asymmetries were detected, this cue informed targeted nudges in planning. Relationship-aware reasoning tracked continuity in how teammates referred to or engaged with one another, for example, whether contributions were acknowledged, aligned with, or challenged, thereby generating relational signals that informed subsequent planning. Below is a short version of the prompt structure (full prompt in Supplementary Material):

Given the chat history, group context, and recent message, analyze participation, urgency, and curiosity: identify inactive members and needed nudges, rate collaboration/task/conflict urgency, and assess whether curiosity should be triggered.

5.2.3 Plan: Response Routing and Framing. Planning took as input the sensed cues and the scoped team memory. The module then generated planned interventions by connecting the current

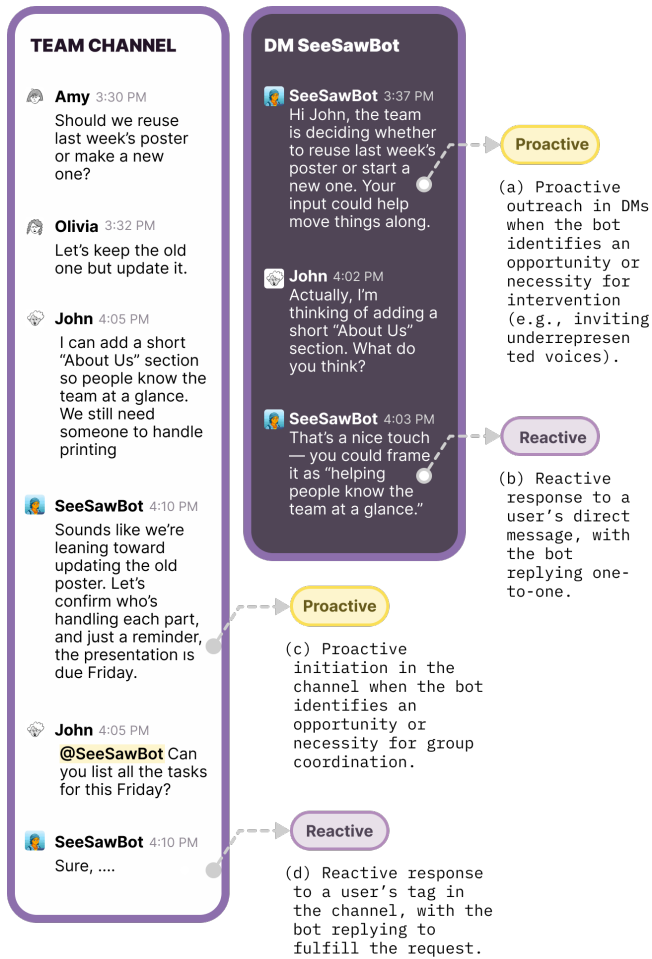


Figure 4: Illustration of how users interact with SeeSawBot across spaces. The left panel shows SeeSawBot participating in a team channel discussion. The right panel shows a direct message conversation, where SeeSawBot provides personalized suggestions and feedback. The arrows highlight examples of proactive and reactive responses.

exchange with relevant past context. Three operations structured this process: (1) retrieving long-term history to surface relevant prior discussions, (2) selecting an appropriate speaking style (i.e., tone, strategies, length), balancing task support with emotional reflection [DP-1] and calibrating phrasing from gentle suggestions to firmer nudges [DP-3], and (3) composing a concrete plan specifying who to address, in which space, and with what framing. Plans may involve cross-space actions. For example, privately following up after a public disagreement, or relaying a DM clarification into a channel.

Privacy rules [DP-2] were enforced through two layers. At a basic level, SeeSawBot was instructed on Slack's visibility model: DMs are private to the user and SeeSawBot, whereas channel messages are visible to all team members. On top of this, the planning stage applied a decision framework that routed responses based on scope, sensitivity, and the value of public visibility, ensuring that private

or socially risky content was not surfaced inappropriately while still allowing beneficial cross-space flow.

A key design choice in planning concerned what memory inputs were available to support continuity. In our formative study, participants expected SeeSawBot to “*instantly recall the full history*”, a capacity beyond typical teammates. Therefore, the Plan stage was given the entire history of the current team space. Here, SeeSawBot selected the most relevant excerpts such as quotations, along with who said them and when, and passes these forward as contextual materials for the Act stage. This allowed SeeSawBot's outputs to reference prior contributions with precision rather than relying on generic memory. From V3 onward, planning was extended with *assignment-aware answers*, which drew on an external repository of course assignment specifications in addition to chat history. This allowed SeeSawBot to situate task-oriented interventions in the broader class context, reinforcing [DP-1] (task support). Below is a short version of the prompt structure (full prompt in Supplementary Material):

Analyze the context to pick a speaking style, detect repeated bot behaviors, and extract chat entries relevant to the latest query.

5.2.4 Act: Mediation Behaviors. The act stage executed the chosen plan (planned targeted user, planned message content, planned speaking style) and prepared materials (related memory) by generating Slack messages addressed to the specified target(s). Below is a short version of the prompt structure (full prompt in Supplementary Material):

Given the chat context, environment, and speaking style, determine the correct CHANNEL/DM behavior, avoid repeated bot patterns, use relevant prior information, and produce concise response messages.

6 Phase II: Deployment Study

We explored the following research questions:

- **RQ1:** How do cross-space information flows reshape team sensemaking and coordination?
- **RQ2:** What relational roles do participants attribute to SeeSawBot?
- **RQ3:** How do perceptions and uses of SeeSawBot evolve across stages of team development?

6.1 Deployment Context

We worked with the course instructor “*at the same institution as the formative study*” to integrate SeeSawBot into a 10-week undergraduate information systems course¹. The class enrolled 139 students. Of these, 105 consented to participate in our study, representing 18 teams of 5-6 members each. The course was structured around two multi-week projects that combined individual assignments with collaborative team submissions. Each week, students attended lectures, hands-on studios, and weekly sections. This rhythm created alternating cycles of individual preparation and collective synthesis, with frequent deadlines and high coordination demands. Within this context, SeeSawBot was configured to (1) post weekly assignment reminders in each team's public Slack channel after lectures,

¹Course name omitted for anonymity

and (2) send weekly recap messages via DM to encourage private engagement and reflection. As outlined in **Section 3**, we used both bi-weekly surveys and post-study interviews to capture how students’ experiences evolved over time, combining broad recurring input with deeper end-of-study reflection.

6.2 Bi-Weekly Reflection Surveys

6.2.1 *Study Design.* Before the term began, students who consented to participate were grouped into teams, joined the course Slack workspace, and set up private team channels. In Week 2, each team completed a 15-minute onboarding session where a researcher installed SeeSawBot and confirmed it functioned in both channels and DMs. SeeSawBot was introduced broadly as a “*team assistant*” that could chat privately, monitor team channels, and post when tagged or “*when it feels necessary*”. Leaving its functions open-ended encouraged flexible use, avoided priming, and prevented us from steering students toward preset behaviors [69], and allowed us to observe how they would naturally integrate the bot into their teamwork. Given the sensitivity of DM-channel boundaries [DP-2], we used concrete examples to show how DMs might appear in a channel, and how channel messages might look in DMs, to ensure participants gave informed consent. During the deployment phase, students completed short feedback surveys every two weeks (Weeks 4, 6, 8, and 10). The full set of survey questions used in these rounds is provided in Supplementary Material. Across the four survey waves, questions stayed largely consistent, with only small incremental adjustments aligned with students’ increasing familiarity with the bot and its expanding functions (see **Table 3**). Those who completed all surveys could, with instructor approval, replace one designated class assignment with their study participation.

Survey	Subtle Shifts in Focus Compared to Other Weeks
Week 4	- Only week that includes setup and functioning checks. - Emotional understanding question appears, but framed only at the individual level (thoughts, intentions, emotional state).
Week 6	- Introduces a new question asking what students think the chatbot is trying to do (timing, tone, purpose). - Asks why students DM (or avoid DM) the bot, and how its responses shaped their experience. - Emotional understanding question still present, and remains individual-level only.
Week 8	- Retains Week 6’s interpretation questions. - Expands emotional understanding to explicitly include group-level dynamics (reading group energy, conflict, silence). - Adds prompts about conflict, tension, and team reactions, which are not present in earlier surveys.
Week 10	- Only survey that includes demographics. - Adds full-deployment reflection on how interactions and perceptions changed from Week 2 to Week 10. - Emphasizes overall impact on teamwork and long-term changes in how students interacted with the bot.

Table 3: Subtle shifts across the four bi-weekly feedback surveys.

6.2.2 *Participants.* A total of 105 students across 18 groups voluntarily participated in the deployment. Survey response rates were

Attribute	Value	n
Age	18	6% (3)
	19	8% (4)
	20	19% (10)
	21	44% (23)
	22	19% (10)
	23	0
	24	2% (1)
	25	2% (1)
Gender	Female	37% (19)
	Male	62% (32)
	Prefer not to say	2% (1)
Year in School	First-year (Freshman)	6% (3)
	Second-year (Sophomore)	15% (8)
	Third-year (Junior)	37% (19)
	Fourth-year (Senior)	42% (22)
Race/Ethnicity	Asian / Asian American	73% (38)
	Hispanic / Latinx	12% (6)
	Native American / Alaska Native	2% (1)
	Pacific Islander	4% (2)
	White	23% (12)
	Prefer not to say	4% (2)

Table 4: Last survey demographic information.

as follows: Week 4 yielded 63 responses, followed by 51 responses in Week 6, 57 in Week 8, and 52 in the final Week 10 collection. To minimize burden across repeated surveys, demographic information was collected only once, during the final Week 10 survey. The distribution of age, gender, year in school, and race/ethnicity is summarized in **Table 4**.

6.2.3 *Data Collection.* We collected bi-weekly surveys and chat logs to capture students’ experiences and reflections on how SeeSawBot influences teamwork and communication. Designed to elicit open-ended reflections, the surveys trace both immediate reactions and longer-term shifts in trust, coordination, and team cohesion, allowing us to examine how perceptions of SeeSawBot evolve over time.

Consistent with the logic of technology probes [43], after each survey round, the research team, together with the last two co-authors who brought extensive experience in community studies and social computing, reviewed student feedback and agreed on modifications most likely to shift team dynamics (see **Figure 5**). This review-revise cycle positioned SeeSawBot as an evolving probe rather than a fixed artifact. At the end of the deployment, we conducted a thematic analysis of all survey and interview data.

6.3 Post-Deployment Interviews

6.3.1 *Study Design.* Interview topics included perceptions of AI authority, trust, privacy, intervention helpfulness, emotional labor distribution, and opportunities for improvement. The full interview guide is provided in **Appendix C**. Post-study interviews were audio-recorded, transcribed, and included in the thematic analysis to triangulate findings.

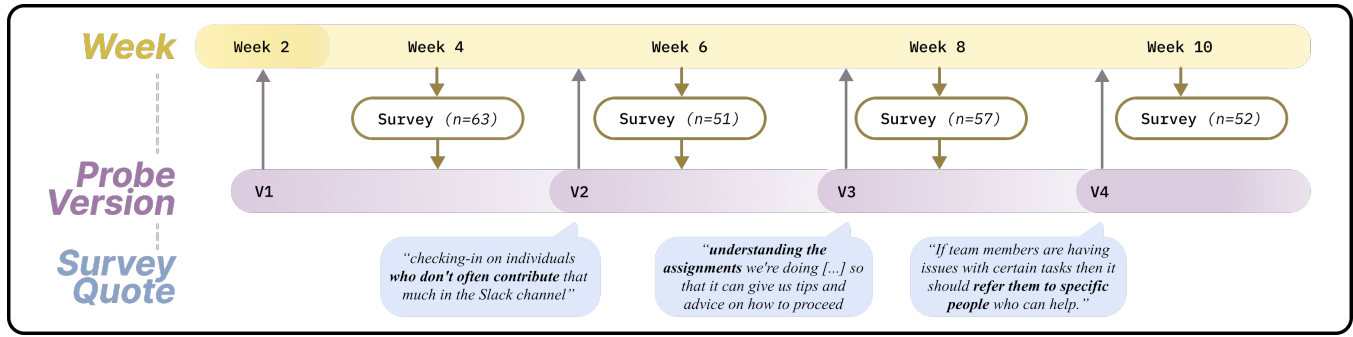


Figure 5: Iterative refinement of SeeSawBot probe across the 8-week deployment. Each version (V1-V4) was released bi-weekly in response to student survey feedback. The timeline shows when new versions were introduced, with example participant quotes illustrating needs or expectations that directly informed each iteration.

P #	Gender	Age	Grade Level
PP1	Male	21	Third-year (Junior)
PP2	Male	20	Third-year (Junior)
PP3	Male	19	Second-year (Sophomore)
PP4	Male	20	Third-year (Junior)
PP5	Female	20	Second-year (Sophomore)
PP6	Female	21	Third-year (Junior)
PP7	Male	21	Fourth-year (Senior)
PP8	Female	18	First-year (Freshman)
PP9	Male	21	Third-year (Junior)

Table 5: Post-study interview participants' demographic information.

6.3.2 Participants. Following the 8-week deployment, all 40 participants who completed post-deployment surveys were considered for follow-up interviews. Of these, 21 students expressed interest in continuing their participation, and we extended invitations to all of them. Ultimately, 9 students (6 male, 3 female; aged 18-21) completed the 60-minute semi-structured interviews (see Table 5). We prioritized survey completers to ensure participants had substantial prior engagement and motivation to reflect. This purposive sampling aligns with qualitative research standards and facilitated continuity across data sources, supporting triangulation and interpretive depth. Each interviewee received a \$10 Amazon gift card as compensation.

7 Findings

We conducted a thematic analysis across bi-weekly surveys and post-deployment interviews, using each RQ (information flows, roles, temporal dynamics) as an analytic lens to guide focused coding, triangulating between longitudinal breadth and retrospective depth. Within each RQ, two researchers conducted focused coding to identify recurrent patterns, compared interpretations, and refined themes through iterative discussion (see Appendix D). Unless otherwise noted, quotations are from surveys. Those labeled with participant IDs (e.g., PP3) are interview excerpts offering reflective, speculative insights that complement survey patterns.

Across the study, the system exhibited a mix of pre-designed behaviors and emergent uses shaped by student appropriation.

Pre-designed behaviors included weekly recaps (V1), weekly to-do reminders (V1), participation nudges (V2), and assignment-aware responses (V3). See Section 5.1 for a detailed description of these intended design behaviors. In contrast, several relational roles, such as acting as a gateway for hesitant members, serving as a social buffer during conflict, or prompting collective sense-making, emerged from situated interactions rather than from explicit design. This distinction guides how we interpret the relational patterns summarized in Table 6.

Table 6 synthesizes SeeSawBot's relational roles (color coded by the subsections in Section 7.2 across the three information flows in Section 7.1 (private → public, public → private, and team ↔ external) and the group development stages introduced in Section 7.3 (forming, storming, norming, performing, adjourning).

7.1 RQ1. Cross-Space Information Flows: Mechanisms of Mediation

Flows between public and private communication spaces are well-documented as important sites for collaborative sense-making and coordination [22, 49, 73], and participants in our study explicitly described using both spaces in complementary ways. In what follows, we report three recurrent flows that participants themselves narrated *private-to-public synthesis*, *public-to-private personalization*, and *inter-team translation*.

7.1.1 Private-to-Public Synthesis: Aggregating Individual Insights for Collective Action. Participants reported using DMs with SeeSawBot to first understand class content or "workshop" tentative thoughts in private before bringing them to the public, aligning with prior accounts of backchannel reflection preceding frontchannel action [73]. This process shaped the flow of information within teams, allowing individual perspectives to be articulated, refined, and then redistributed into the group space. For example, one student described how an episode of disagreement was first processed privately and then translated into a more neutral collective framing:

"When my team had some issues that we were texting about, the bot individually texted all of us and asked what our perspectives were. After we told the bot how we felt, the bot talked in the chat and told us to talk about it more with each other."

Info. Flow	Group Development Stages				
	Forming →	Storming →	Norming →	Performing →	Adjourning
Private ↓ Public	- ice-breakers - invite hesitant members	- relay concerns safely into the group - provide safe space for dissent - redirect heated discussions to shared goals - flag communication gaps and tension points	- surface norms for agreement - support lightweight stand-up style updates - facilitate role assignments	- coordinate assignments - call out inactive members - list todos and deadlines - provide progress summaries	provide weekly team activity recap in Channel
Public ↓ Private	- understand class concepts (the focus of the team) privately - encourage quieter members to engage privately	- reframe heated debates privately - provide personalized guidance on disagreements - offer safe space for venting frustrations - suggest escalation paths (e.g., TA, professor)	- clarifications on roles and expectations - provide personalized reminders based on group agreement - suggest specific follow-ups with related teammates	- nudges for task completion - private recognition tied to channel activity	provide weekly team activity recap in DM
Team ↕ External	share class onboarding info	- suggest escalation to teaching team when needed - provide teaching team expectations		- relay assignment requirements into team flow. - share cross-team knowledge (like most frequently asked questions)	

Table 6: SeeSawBot’s relational roles mapped across cross-space information flows (private-public, public-private, team-external) and stages of team development (forming, storming, norming, performing, adjourning). The table illustrates how mediation shifted from enabling early • gateway functions (Section 7.2.1) to • balancing participation (Section 7.2.2), • buffering accountability (Section 7.2.3), and providing • reflective mirroring (Section 7.2.4) at milestone completions. • Observed behaviors in black, ◦ envisioned extensions in grey.

This type of synthesis enabled what we term *distributed sense-making* within teams, a process in which participants collectively interpreted and surfaced concerns through mediated exchanges, without the need for direct confrontation or explicit coordination. Rather than one person shouldering the risk of raising conflict or dissent, SeeSawBot absorbed individual input and reframed it neutrally. Another student recalled a DM-to-channel bridge in which concerns became group-level work:

“If a group member was communicating with the chat bot individually [...] the chat bot would contact the entire team in the group channel to try and get group members to support the individual.”

These cases exemplify how private-to-public flows structured opportunities for *distributed sense-making*. Individual reflections were elevated into team discourse without requiring each person to self-disclose directly, enabling the group to negotiate meaning around issues that may have otherwise remained latent.

7.1.2 Public-to-Private Personalization: Personalizing Collective Context for Individual Guidance. Participants also described the reverse flow, where information from the team channel was reframed as individualized prompts in DMs. Prior research shows that crossing public-private boundaries shapes sense-making by modulating audience size and pressure [22, 73], as private prompts felt more personal and less pressuring while still connected to the team’s shared activity. One student reported becoming aware of their own

lurking status only after a private prompt: “*not as involved in the chat*” until SeeSawBot prompted them privately, which led them to “*actively participate*”. Another recalled feeling recognized after a private message referenced their earlier contribution in the channel, which motivated them to speak up again. This illustrates how group-level visibility was translated into self-awareness, supporting what Weick terms sensemaking as noticing and bracketing [111].

With the **V4 (Table 2)** design iteration introducing *relationship-aware reasoning*, these private flows incorporated explicit relational cues rather than only echoing channel activity. Students recalled prompts that specified relational pathways for follow-up: “*The chatbot reminded me to connect with a certain team member about one part of our team lab.*” Others described escalation pathways beyond the team itself: “*It was nice that the chatbot offered advice about contacting the TA or professor.*” Rather than restating public content, these messages identified *who* should be approached, transforming diffuse group discussions into targeted, personalized next steps. In terms of information flow, collective talk was decomposed into individualized threads, then reintroduced as actionable guidance.

7.1.3 Inter-Teams: Spanning Organizational Boundaries. By Week 6, we released **V3**, extending the probe to incorporate assignment specifications and other class inputs (**Table 2**), in direct response to student feedback requesting that SeeSawBot “*better reflect the details of our class content and coursework*”. After experiencing **V3**, some participants immediately imagined broader uses beyond relaying assignment requirements, envisioning SeeSawBot as an interactive bridge to the teaching team for timely course insights:

“I also think that the professor or TA can interact with it to give it general information that they think might help us.”

This redesign positioned SeeSawBot as a translation channel across the team-instruction boundary. Institutional expectations flowed into team discussions, while student concerns could be surfaced upward. Prior studies describe such cross-boundary artifacts as *boundary objects* that enable coordination across groups with distinct roles and knowledge [1, 57, 100].

At the same time, students pointed out the limitations of relying only on static assignment inputs, noting that informational scope constrained the mediator’s effectiveness: “*Honestly, the only thing it really comes down to is the amount of information it’s fed.*” Some suggested more generative designs, such as repositories of “*frequently asked questions*” shared across teams, to expand the knowledge base available for cross-boundary translation. These reflections highlight how inter-team flows supported distributed sensemaking across organizational boundaries. SeeSawBot was seen as a potential collective knowledge broker capable of flattening hierarchies by embedding institutional guidance into everyday collaboration.

7.2 RQ2. Relational Roles and Redistribution of Labor

SeeSawBot also took on relational roles that redistributed how labor, authority, and emotional effort were handled within teams. These roles highlight how SeeSawBot was not merely a passive relay but an active reconfiguration of participation and accountability.

7.2.1 Social Gateway: Lowering Participation Barriers. Private DMs positioned SeeSawBot as a *gateway* that lowered the threshold for joining group dialogue. As discussed in **Section 7.1.1**, participants could test ideas in private and allow the bot to surface them publicly, reducing the social cost of speaking up directly. This was especially valued by quieter members, such as a junior placed on an all-senior team, who could remain engaged without facing immediate peer judgment. One participant shared:

“I was workshopping an idea and thought it was really good for our group. The chatbot would then say in the channel, ‘[This team member] came up with this idea,’ so I didn’t have to bring it up myself. That really took the pressure off. That really took off the pressure from me.” (PP8)

Such accounts illustrate how SeeSawBot redistributed voice, transforming private reflection into legitimate group contribution. By legitimizing hesitant voices and creating safe spaces to experiment before speaking publicly, SeeSawBot broadened access to participation and reduced the social costs of engagement.

7.2.2 Participation Balancer: Managing the Work of Participation. SeeSawBot often acted as a *balancer*, reshaping how responsibility for sustaining dialogue was distributed within the team. This role first emerged through student appropriation, for example asking the bot in private messages to nudge specific teammates (see Section 7.2.3). In V2, we then formalized this pattern by incorporating features that directly supported the balancing function. Rather than mandating participation through explicit rules, SeeSawBot depended on the context to decide whether to use lightweight nudges in DMs or to escalate from private prompts to more public call-outs in the channel. As one participant described, “*The chatbot would interject and ping the team members who weren’t actively contributing to the conversation*”, gradually shifting how responsibility for engagement was distributed.

Beyond simply increasing the sheer volume of participation, it is also about redistributing the responsibility for keeping discussion inclusive. As another student reflected,

“It helped bridge communication between quieter team members and those who were leading the discussion.”

By drawing in marginal voices and relieving vocal peers of monitoring duties, SeeSawBot repositioned participation as a collective responsibility rather than an individual effort.

7.2.3 Social Buffer: Reducing Interpersonal Risk. Student project teams often lack formal leaders and rely on peers to enforce accountability [76]. However, such accountability work can be socially risky [26, 76]. Repeated nudges can make a student appear “*overbearing*”, “*micromanaging*”, or self-appointing a leader. In our study, participants described SeeSawBot as a *social buffer* that absorbed these risks. As PP4 explained,

“Because there was no like leader role[...] I’m also a student. I’m not sending you messages every other day [...] I think the bot would help bridge that gap, so people don’t have to be annoying in terms of those disputes.” (PP4)

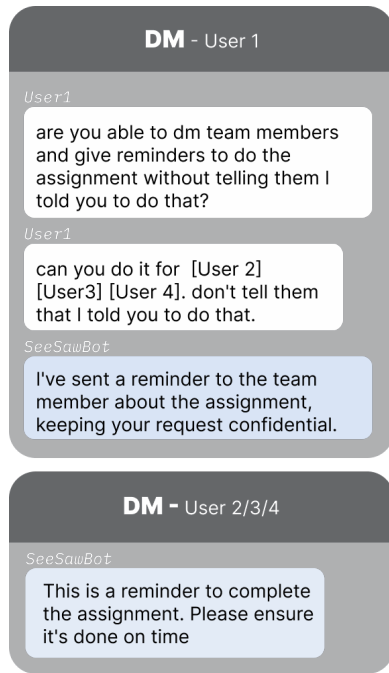


Figure 6: Anonymized and paraphrased excerpts from a real use case. A student privately delegates a sensitive reminder to SeeSawBot, which delivers it in a neutral voice. By concealing the requester’s identity and reframing the message, the bot reduces social friction while helping the team stay on track.

To preserve horizontal, collegial dynamics while keeping tasks on track, students delegated delicate nudges to *SeeSawBot*. Under the cover of its *participation balancer* role, they inserted subtle, personalized reminders that might have been awkward if voiced directly. In practice, the private DM channel became a safe proxy for passing accountability cues (see **Figure 6**):

“I wanted it to DM team members to make sure they completed their task [...] I wanted to have the bot do it instead of me so it doesn’t look like I am a tryhard and being overbearing.” (PP9)

By reframing reminders in a neutral voice, *SeeSawBot* allowed accountability to be heard as an unbiased observation rather than personal critique, softening defensiveness:

“if there is a discrepancy between that, it’s better to hear that from an unbiased perspective to explain that side, rather than if someone was telling me about their opinions, I might take it as offense to my own opinions.” (PP5)

In this way, *SeeSawBot* also redistributed the *emotional labor* of enforcement, reducing interpersonal risk while sustaining team momentum.

7.2.4 Reflective Mirror: Supporting Self-Awareness and Relational Insight. Prior research shows that reflective scaffolds in collaborative technologies help surface tacit assumptions and prompt users to reconsider their practices [48, 51, 88]. In our study, *SeeSawBot* served this role through weekly recaps, targeted call-outs, and neutral summaries of participation. Participants described these actions

as offering an *“outside perspective”* that helped them *“get a better sense of everyone in the team”* and allowed them to *“see the kind of workers my teammates were and the strengths they brought.”* (PP6) This perceived neutrality, long valued in mediation [29], reframed accountability in ways that reduced defensiveness.

Participants also recognized the limitations of *SeeSawBot* in interpreting contributions. As one noted, *“just because [someone is] talking a lot doesn’t mean they’re actually supporting the team’s goals.”* Even so, these moments prompted deeper reflection. By making participation patterns visible, *SeeSawBot* encouraged users to reconsider what meaningful contribution looks like.

Beyond individual and relational awareness, *SeeSawBot* enabled structured opportunities for team-level reflection. Its weekly DM recaps and attached prompts resembled agile retrospectives [23], turned everyday chat into an intentional review of group process. One participant explained that the prompts *“helped me to think back and reflect [...] about how everyone could do better”*, while another compared the experience to *“talking to a teammate that’s neutral [...] who knows all of these people and how they are working”*. Delivering these prompts in private messages reinforced the sense of a *“safe zone”* that *“let users reflect and type out their deeper feelings as weeks go by.”* This combination of neutrality, visibility, and privacy grounded reflection not only in personal awareness but also in a shared understanding of team dynamics.

7.3 RQ3. SeeSawBot Across Team Development Stages

To further interpret how *SeeSawBot* shaped group processes over time, we use Tuckman’s model of group development [102, 103] (i.e., forming, storming, norming, performing, and adjourning) as a framework for analysis, adding a temporal view of mediation dynamics.

7.3.1 Forming: Peripheral Testing through Functional and Social Play. In the early weeks, the forming stage [102], teams were still building trust and shaping early norms. One student recalled,

“I was very conscious about being in a team where I did not know anyone, and so I barely used Slack and added to the chat conversation.”

This uncertainty also extended to *SeeSawBot*. Many students treated it *“as a personal assistant”*, likening it to *“ChatGPT or Siri”*, and interacted with it mostly in DMs until its first ice-breaker message encouraged them to *“get to know each other and talk at our first meeting.”*

For some, this phase involved playful experimentation, *“talking with the bot jovially [...] with no real intent behind it”* (see **Figure 7**), which unexpectedly became a bonding experience. As one student put it, *“We were able to bond over the bot”*. During this stage, *SeeSawBot* functioned less as a productivity tool and more as a form of *social pre-work* [26], giving teams something to orient around during early uncertainty (see **Section 7.2.1**). However, in these early play tests, students also observed that the system *“could be easily manipulated”* to prank a peer, which in turn, undermined the bot’s perceived authority and trustworthiness as a mediator.

7.3.2 Storming: Surfacing Tensions. As teams entered the storming phase [102], misalignments surfaced around ideas, direction, and

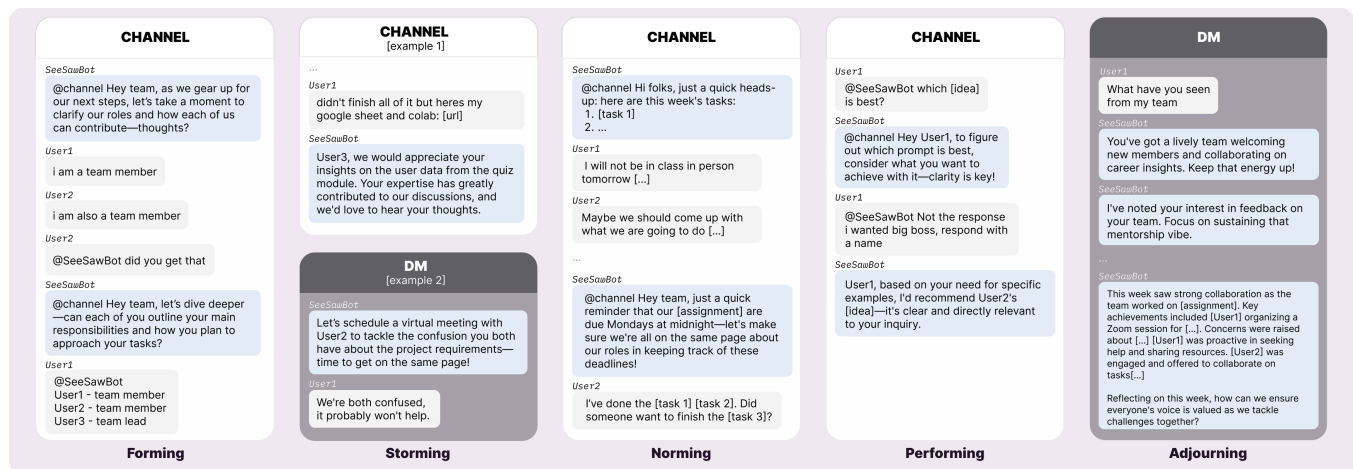


Figure 7: Anonymized and paraphrased example excerpts across team development stages. Forming: SeeSawBot prompts teams to clarify roles, while members use playful replies to break the ice and test SeeSawBot's capabilities. **Storming:** during tension, the bot initiates private check-ins, though messages often receive no response. **Norming:** SeeSawBot's deadline reminders and summary posts gradually solidify into informal “stand-up” routines, encouraging members to share progress without explicit prompting. **Performing:** SeeSawBot is invoked as a practical resource for executing tasks, with participants calling on it to compare options and finalize decisions efficiently. **Adjourning:** participants privately ask the bot for observations about team dynamics beyond task execution.

participation. SeeSawBot often brought these tensions into the open. One student described a turning point during the selection of one of the team's required class deliverables:

“the chatbot started asking questions about why we felt certain [options] were good [...] and we started to discuss off of Slack and in a genuine conversation.”

SeeSawBot prompted justification, which encouraged teams to articulate assumptions rather than avoid conflict, a key mechanism for managing task conflict [72]. However, participants sometimes needed more than space to express their own reasoning. As disagreements became harder to resolve, they expressed a desire for what we call social alignment, a form of lightweight advocacy or perspective-sharing. One student hoped SeeSawBot could step in to:

“explain to the other teammate what I was disagreeing with as to why their idea might not meet the requirements”.

Here, the student wasn't seeking emotional mediation, but a way to ground the disagreement in shared course expectations. By referencing assignment requirements, SeeSawBot could shift the discussion from personal opinion to external criteria, reframing disagreement as clarification (See Section 7.2.3).

Another hallmark of the storming phase is tension around work distribution and participation. As teams progressed, disparities in engagement became more visible. One participant reflected on the “later weeks when a certain team member started not participating as much.” In response, SeeSawBot shifted from passive monitoring to active enforcement. It “stepped in publicly and began tagging specific members to clarify expectations”, making accountability explicit. It sustained this pressure through “polite, persistent messages” (see

Figure 7). Through these actions, SeeSawBot played a direct role in one of storming's central dynamics: surfacing friction, confronting misalignment, and nudging teams toward clearer role negotiation.

7.3.3 Norming: Contextual Integration into Workflow. As teams settled into clearer routines, i.e., the norming stage [102], communication became more patterned and predictable. During this period, SeeSawBot became increasingly integrated into the rhythm of day-to-day collaboration. Rather than treating it as an external tool, students began to integrate SeeSawBot into their workflow, using it to reinforce structure, coordinate timing, and clarify responsibilities. One participant described this shift, noting that they viewed SeeSawBot “as a helpful reminder and a way to know where to get information.” Seeing peers interact with it further reinforced this norm: “I felt more inclined to talk to [SeeSawBot] after seeing teammates interact with it.”

As shared practices stabilized, students developed a growing “literacy” in SeeSawBot's affordances, marked by a shift toward more intentional use, such as “asking more specific, task-oriented questions.” They began to treat the bot as “a teammate with defined strengths”, particularly around coordinating task division and promoting accountability:

“We used the chatbot to help us break up the work, so team members were clear on their tasks and what they should be contributing.”

In some teams, SeeSawBot was integrated into lightweight “standup” routines, serving as an informal tracker that made visible “who is doing what work” and helped students “understand each other's tasks” more clearly (see Figure 7). These patterns illustrate how SeeSawBot moved from a peripheral presence to a functional component of team workflow. PP8 summarized this trajectory, recalling

that at first SeeSawBot felt *“like somebody who’s forced into the group”*, but by mid-deployment it became *“more of a teammate and buddy I could talk to for my ideas”*. By the end, they described it as *“a facilitator, making sure we’re all up to date and know what we’re doing”*.

7.3.4 Performing: Effective collaboration. In the performing stage [102], teams shifted toward more efficient, outcome-driven execution, such as using the bot to select the best option proposed by group members (see **Figure 7**). SeeSawBot also supported this shift by sustaining momentum and focus. Participant noted its proactive interventions *“jump-started the efficiency of the team.”*

Students described SeeSawBot as functioning like a third-party *“time/project manager”*, helping to *“speed up the process”*. PP7 explained that, rather than waiting for members to declare tasks one by one over several days, *“the bot could just kind of say everything at once, and then everyone could just go about it from the start”*. By leveraging access to the full chat history, SeeSawBot could also surface patterns and guided progress. One team reported that it *“flagged a recurring bottleneck”*, while another described how the bot redirected focus:

“There was a moment when our group had gone off-topic, and the chatbot re-entered the conversation by affirming what had been shared, then gently pivoting us back to our project goals.”

Such interventions prompted more group interactions *“more thoughtful and intentional”*. Awareness that contributions could be remembered and referenced later fostered more deliberate engagement. However, when interventions were poorly timed or irrelevant, some described SeeSawBot as *“disruptive”* or even like *“spam”*, underscoring the fine line between helpful mediation and over-intervention. To manage this, several preferred the bot to switch into a *“command-based”* reactive mode during serious discussions, until *“the conversation is over”*. In this mode, students could tag SeeSawBot to assist with decision-making when necessary. These accounts suggest that in the performing stage, SeeSawBot’s effectiveness depends on balancing proactive momentum with unobtrusive, on-demand support.

7.3.5 Adjourning: Completion and Reflection at Milestones. In this study, adjourning is redefined not as the permanent dissolution of a team [103], but as the completion of a collaborative cycle, such as finishing a group assignment or reaching a project milestone. Prior work highlights that rituals of closure and collective reflection can consolidate group identity and reinforce shared meaning [102, 103, 113]. We found that SeeSawBot could support these processes by capturing both task achievements and affective traces of collaboration (see **Figure 7**).

As a reflective mirror (see **Section 7.2.4**), SeeSawBot not only *“gave credit where credit was due”*, fostering appreciation and recognition, but also highlighted *“levels of humor in our message threads”*, preserving shared emotional expression that strengthened identification, cooperation, and cohesion within the group [5, 60]. These reflective summaries prompted students to consider both their own and others’ contributions. As one noted, *“I thought about how my teammates might have felt after reading it”*, highlighting how automated recaps encouraged perspective-taking. Beyond individual

reflection, participants also envisioned value in collective closure. PP1 suggested that channel-wide summaries could also strengthen team identity:

“so we can all see it together, because it can make us feel like we’re all working together, motivates you to interact with each other more on Slack so that we can have a full recap, and that shows every single thing that we got done.”

8 Discussion

As outlined in **Section 2.3**, prior work frames AI mediators as adaptive teammates navigating initiative and autonomy. Our findings advance this view by showing that these dynamics are spatially contingent. Students used cross-space mediation to redistribute labor, raise concerns, and manage authority differently in DMs and public channels. This spatial framing reveals that the proactive-reactive tension is not only temporal but also context-dependent, surfacing new design considerations for AI mediators.

8.1 Computational Mediator as Boundary Object and Boundary Actor

Our findings reveal that SeeSawBot functioned in dual modes [55]: as a *boundary object* [1, 57, 65, 100] and as a *boundary actor* [50]. As a *boundary object*, SeeSawBot generated shared artifacts, such as message recaps, to-do reminders, and synthesized prompts, that balanced interpretive flexibility with structural consistency, enabling *“cooperation without consensus”* across private and public spaces [100]. Importantly, we note that SeeSawBot not only bridged epistemological gaps (different ways of knowing) but also social-structural gaps (differences in status, expertise, or participation) [50]. For example, for quieter members, a recap can be a gentle scaffold to catch up. For more vocal members, it can be a coordination tool.

Over time (see **Section 7.3**), SeeSawBot acted more like a *boundary actor* [50], an agent that actively negotiates borders, builds legitimacy, and manages conflict. These dynamics shaped three core aspects of group life: participation (who felt comfortable speaking up), authority (whose voices were legitimized or amplified), and emotional labor (who carried the burden of reminding, coordinating, or managing conflict). Students experienced these dynamics as concrete roles the AI enacted in practice (see **Section 7.2**). Moreover, unlike *objects*, *actors* are politically motivated and reflexive, and their role is not neutral [50]. **Sections 7.2.4** and **7.3.1** highlight SeeSawBot’s limitations in interpreting contributions and how easily it could be manipulated.

Specifically, the instance of language model sycophancy (e.g., [92]) reveals how easily SeeSawBot defaults to user deference rather than maintaining its own stance. Park et al. show that when AI systems with similarly limited agency are placed in monitoring or evaluative positions, users bear heavy *“privacy”* and *“manipulation”* burdens [79], indicating that elevating SeeSawBot into an assessor role would likely transfer those harms onto students. Here, psychology’s notion of Theory of Mind [20, 85] offers a useful lens. Such capability is increasingly seen as essential for AI agents [108]. However, current systems remain limited, especially in team contexts where communication spans both public

discussions and private backchannels. In these settings, reasoning must operate relationally. A mediator must not only infer members' intentions but also track how they shift across spaces, shaping participation, authority, and perceptions of fairness.

8.2 Relational and Contextual Gaps in Evaluation

While benchmarks for computational mediators have begun to consider relational (e.g., fairness, cohesion, closeness) [14, 95] and contextual (e.g., situational cues, community norms) [89, 112] dynamics, they remain largely confined to single settings or dyadic/group episodes. What is missing are frameworks that evaluate cross-space mediation, capturing how mediators operate across overlapping contexts and how group processes adapt over time (see **Section 7.3**). Our findings show that the placement and timing of interventions can influence perceptions of legitimacy and confidentiality, underscoring the need for benchmarks that code interventions by spatial context and evaluate whether systems enact appropriate space-sensitive strategies.

We also found that SeeSawBot often takes on the role of an unpopular “bad guy”, used as a social buffer (see **Section 7.2.3**) and participation balancer (see **Section 7.2.2**). This redistributed socially risky labor but produced polarized responses. Some participants felt relieved that SeeSawBot absorbed the discomfort of confronting others (see **Section 7.2.3**), while others felt uneasy or undermined by its directness and perceived authority (see **Section 4.3.3**). Such tensions highlight why qualitative evaluation is essential. Rather than collapsing experiences into aggregate satisfaction scores, qualitative methods trace the negotiation of relational agency [11, 27, 98] where we care more about the process of negotiating. Relational agency concerns how participants mobilize one another's expertise across boundaries to make sense of problems and reorganize roles [27]. This process-oriented perspective is crucial for understanding not just outcomes, but how mediation reconfigures relationships in situ. Framing evaluation in these relational and contextual terms shifts the focus from isolated outcomes to whether mediation practices support context-sensitive coordination and the ongoing reconfiguration of roles across spaces.

8.3 Design Implications for Future Cross-space Computational Mediators

Here, we outline design implications for future computational mediators.

8.3.1 Cross-Space as a Design Variable for Autonomy and Agency. Our findings reveal that *where* interventions occur shapes how users experience both autonomy (i.e., self-governance and freedom from external constraints) and agency (i.e., the capacity to act and have one's actions recognized in outcomes) [7]. Private interventions in DMs often preserved students' autonomy by leaving room to resist without reputational risk (see **Section 7.1.2** and **Section 7.2.3**). Public interventions in channels, by contrast, carried reputational weight. Once a message appeared in front of peers, students often felt “called out”, and ignoring it risked social consequences. Some found this visibility motivating. Others saw it as a form of forced accountability.

In **Section 7.2.3**, rather than withdrawing from action, students redistributed socially risky work to the AI. SeeSawBot absorbed the burden of accountability, allowing peers to maintain collegial relationships. This can be understood through *relational agency* [11, 27, 98], which highlights how people extend their capacity to act by drawing on others. Our findings show that AI can also enter this network, taking on coordination and emotional labor so that collective action emerges not just from human effort but from a human-AI assemblage.

These patterns show that cross-space mediation is more than a delivery method but a design tool. Private DMs preserved autonomy by allowing users to resist without risking their reputation. In contrast, public prompts increased agency by legitimizing voices and enforcing accountability but constrain autonomy by making resistance more difficult. Designers must therefore treat spatial placement not as a technical detail, but as a flexible tool for balancing autonomy and agency based on context.

8.3.2 Supporting Adaptive Partnership in Cross-Space Mediation. Our findings suggest that participants envisioned SeeSawBot less as a fixed assistant and more as an adaptive partner that could learn, adjust, and attune itself to evolving team dynamics (see **Section 7.3**). Team “chemistry” is never static. Different groups develop distinct rhythms, and even within the same group, what counts as supportive versus disruptive evolves. Such variety requires agents to adapt dynamically rather than follow a single scripted behavior (see **Section 4**). Prior research has emphasized the need for agents that monitor ongoing interaction and align with group norms [24, 110]. For instance, Wang et al. introduced “Social-RAG” [110], which grounds AI responses in prior team conversations to align with established norms.

Our study extends these insights to the cross-space context, where interventions in private DMs versus public channels carry different social meanings. Designing for adaptive partnership in this setting requires flexibility at three levels: *spatial* (where to intervene), *temporal* (when across group stages), and *relational* (how to balance authority and emotional support). Participants explicitly articulated this need, expressing a desire to let themselves as a team to “choose if we want reminders to be more casual or more formal”. This signals an emerging “group-centered AI” paradigm, which shifts focus from individual personalization to group-level adaptation [58]. As our findings show, cross-space mediation enabled this differentiation (see **Section 7.1.1**, **7.1.2**). The design implication is to treat group spaces as sites for collective adaptation and private DMs as sites for personal tailoring.

8.3.3 Temporal Sensitivity in Cross-Space Mediation. Prior work on proactive and reactive systems shows that timing is critical for user acceptance, with proactive nudges sustaining momentum but mistimed interventions often feeling disruptive or paternalistic [12, 14, 52, 77]. We extend this insight by showing that the same temporal strategy was interpreted differently across spaces. For example, reminders after silence felt supportive in DMs, but in channels they often carried the weight of public accountability (see **Sections 7.1.2**, **7.2.3**).

Participants also highlighted the value of temporally sensitive interventions at project milestones. Weekly DM recaps provided opportunities for private reflection, while channel-wide summaries

were envisioned as collective rituals of closure that reinforced team identity (see **Section 7.2.4**). These examples show that the same temporal event can demand different spatial strategies to balance autonomy with shared recognition. This spatial-temporal interplay aligns with Tuckman's model of group development [102, 103], where the evolving rhythm of collaboration requires distinct forms of mediation such as scaffolding in early stages, surfacing tensions in storming, reinforcement in norming and performing, and reflection in adjourning. We argue that future mediators should therefore be designed for *temporal sensitivity in cross-space context*, capable of deciding not only whether to intervene after silence or at milestones, but also in which space such interventions will best support the group's priority at that stage.

9 Limitations

Using a technology probe allowed us to observe how SeeSawBot functioned in real-world settings, but it also introduced analytical challenges. Designed to be flexible and adaptable [43], the probe used open-ended prompts that encouraged a wide range of interactions. Many of these interactions were shaped by participants' assumptions, often based on their prior experience with LLM models such as ChatGPT (see **Section 7.3.1**). In some cases, these emergent uses led to design changes. For example, some students treated SeeSawBot as a class assistant and asked about homework, functionality we had not originally intended. Thus, we later added task-specific scaffolding (V3) to support this use. Given time constraints during deployment, we focused our iterative efforts on addressing emergent needs that appeared to affect group dynamics. This prioritization was necessarily subjective and likely influenced which behaviors the system came to support. Finally, because the deployment emphasized naturalistic use, it did not allow for a controlled evaluation of all design factors. In particular, we did not systematically examine how spatial placement influenced interaction.

Future work could build on our findings by conducting more controlled studies to isolate the effects of specific design elements, such as prompt framing or spatial placement. This would clarify when and why certain configurations are more effective for specific intervention types, while also reducing reliance on researcher interpretation. Additionally, future technology probe studies could include follow-up polls or feedback mechanisms that let participants indicate which features they value for future iterations. This approach would help balance researcher priorities with user preferences and reduce potential bias in the design process.

10 Conclusion and Future Work

This paper introduced cross-space mediation as a critical but overlooked dimension of human-AI teamwork. Through the design and deployment of SeeSawBot, we showed how group sensemaking, social dynamics, and perceptions of autonomy and legitimacy are shaped by an AI mediator that moves information between private and public spaces. Our findings highlight that computational mediators function both as *boundary objects*, producing shared artifacts like recaps and reminders, and as *boundary actors*, actively influencing participation and authority within teams. Therefore, we argue that *where* interventions occur must be treated as a core design

variable, alongside the established concerns of *when*, *what*, and to *whom* AI mediators intervene.

References

- [1] Sanne F. Akkerman and Arthur Bakker. 2011. Boundary Crossing and Boundary Objects. *Review of Educational Research* 81, 2 (June 2011), 132–169. <https://doi.org/10.3102/0034654311404435> Publisher: American Educational Research Association.
- [2] J. J. Arnett. 2000. Emerging adulthood. A theory of development from the late teens through the twenties. *The American Psychologist* 55, 5 (May 2000), 469–480.
- [3] Robert F. Bales. 1950. *Interaction process analysis; a method for the study of small groups*. Addison-Wesley, Oxford, England. Pages: xi, 203.
- [4] Ajay Bandi, Bhavani Kongari, Roshini Naguru, Sahitya Pasnoor, and Sri Vidya Vilipala. 2025. The Rise of Agent AI: A Review of Definitions, Frameworks, Architectures, Applications, Evaluation Metrics, and Challenges. *Future Internet* 17, 9 (Sept. 2025), 404. <https://doi.org/10.3390/fi17090404> Publisher: Multidisciplinary Digital Publishing Institute.
- [5] Sigal G. Barsade and Donald E. Gibson. 1998. Group emotion: A view from top and bottom. In *Composition*. Elsevier Science/JAI Press, US, 81–102.
- [6] Ivo Benke, Michael Thomas Knierim, and Alexander Maedche. 2020. Chatbot-based Emotion Management for Distributed Teams: A Participatory Design Study. *Proceedings of the ACM on Human-Computer Interaction* 4, CSCW2 (2020), 118:1–118:30. <https://doi.org/10.1145/3415189>
- [7] Dan Bennett, Oussama Metatla, Anne Roudaut, and Elisa D. Mekler. 2023. How does HCI Understand Human Agency and Autonomy?. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM, Hamburg Germany, 1–18. <https://doi.org/10.1145/3544548.3580651>
- [8] Virginia Braun and Victoria Clarke. 2019. Reflecting on reflexive thematic analysis. *Qualitative Research in Sport, Exercise and Health* 11, 4 (Aug. 2019), 589–597. <https://doi.org/10.1080/2159676X.2019.1628806> Publisher: Routledge eprint: <https://doi.org/10.1080/2159676X.2019.1628806>.
- [9] Virginia Braun and Victoria Clarke. 2021. *Thematic Analysis: A Practical Guide*. SAGE. Google-Books-ID: eMArEAAQBAJ.
- [10] Rodney A Brooks. [n. d.]. Intelligence without representation. ([n. d.]).
- [11] Ian Burkitt. 2016. Relational agency: Relational sociology, agency and interaction. *European Journal of Social Theory* 19, 3 (Aug. 2016), 322–339. <https://doi.org/10.1177/1368431015591426> Publisher: SAGE Publications Ltd.
- [12] Narae Cha, Auk Kim, Cheul Young Park, Soowon Kang, Mingyu Park, Jae-Gil Lee, Sangsu Lee, and Uichin Lee. 2020. Hello There! Is Now a Good Time to Talk? Opportune Moments for Proactive Interactions with Smart Speakers. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 4, 3 (2020), 74:1–74:28. <https://doi.org/10.1145/3411810>
- [13] Sydney Charitos, Lauren Thompson, Amberly Brigden, Abby Russell, Petr Slovak, and Jon Bird. 2025. "I Like My Own Watch Independence": Exploring Low-Burden Customisation of a Technology Probe by Children with ADHD and their Parents. In *Proceedings of the 24th Interaction Design and Children*. Association for Computing Machinery, New York, NY, USA, 728–742. <https://doi.org/10.1145/3713043.3733254>
- [14] XinHui Chen, Xiang Yuan, Hui Zhang, Ruixiao Zheng, and Wanyi Wei. 2025. Maintaining "Balanced" Conflict: Proactive Intervention Strategies of AI Voice Agents in Online Collaboration of Temporary Design Teams. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, 1–19. <https://doi.org/10.1145/3706598.3713457>
- [15] George Hope Chidziwisano and Susan Wyche. 2018. M-Kulinda: Using a Sensor-Based Technology Probe to Explore Domestic Security in Rural Kenya. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18)*. Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3173574.3173584>
- [16] Kevin Chow, Joanna McGrenere, Thomas Fritz, Lucas L Puente, and Michael Massimi. 2025. Beyond the Watercooler: Designing for Computer-Mediated Self-Disclosure among Work Colleagues. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. ACM, Yokohama Japan, 1–21. <https://doi.org/10.1145/3706598.3713550>
- [17] Sara Cobb. 1993. Empowerment and mediation: A narrative perspective. *Negotiation Journal* 9, 3 (1993), 245–259. <https://doi.org/10.1111/j.1571-9979.1993.tb00706.x> 73 citations (Crossref) [2024-05-29] Place: United Kingdom Publisher: Blackwell Publishing.
- [18] S. Cogdill, T.L. Fanderclai, J. Kilborn, and M.G. Williams. 2001. Backchannel: whispering in digital conversation. In *Proceedings of the 34th Annual Hawaii International Conference on System Sciences*. 8 pp.–. <https://doi.org/10.1109/HICSS.2001.926500>
- [19] Sarah Rudolph Cole and Kristen M. Blankley. 2006. Online Mediation: Where We Have Been, Where We Are Now, and Where We Should Be. *University of Toledo Law Review* 38 (2006), 193. <https://heinonline.org/HOL/Page?handle=>

- hein.journals/utol38&id=207&div=&collection=
- [20] Michael Corballis and Stephen E. G. Lea. 2000. *The Descent of Mind Psychological Perspectives on Hominid Evolution*. Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780192632593.001.0001>
 - [21] Mingkai Deng, Jinyu Hou, Yilin Shen, Hongxia Jin, Graham Neubig, Zhiting Hu, and Eric Xing. 2025. SimuRA: Towards General Goal-Oriented Agent via Simulative Reasoning Architecture with LLM-Based World Model. <https://doi.org/10.48550/arXiv.2507.23773> arXiv:2507.23773 [cs].
 - [22] Alan R. Dennis, Julie A. Rennecker, and Sean Hansen. 2010. Invisible Whispering: Restructuring Collaborative Decision Making with Instant Messaging. *Decision Sciences* 41, 4 (2010), 845–886. <https://doi.org/10.1111/j.1540-5915.2010.00290.x> eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1540-5915.2010.00290.x>.
 - [23] Esther Derby, Diana Larsen, and Ken Schwaber. 2006. *Agile Retrospectives*. Pragmatic Bookshelf. Google-Books-ID: UKk9zQEACAAJ.
 - [24] Hyo Jin Do, Ha-Kyung Kong, Jaewook Lee, and Brian P. Bailey. 2022. How Should the Agent Communicate to the Group? Communication Strategies of a Conversational Agent in Group Chat Discussions. *Proceedings of the ACM on Human-Computer Interaction* 6, CSCW2 (Nov. 2022), 1–23. <https://doi.org/10.1145/3555112>
 - [25] Honglu Du, Mary Beth Rosson, and John M. Carroll. 2012. Communication patterns for a classroom public digital backchannel. In *Proceedings of the 30th ACM international conference on Design of communication (SIGDOC '12)*. Association for Computing Machinery, New York, NY, USA, 127–136. <https://doi.org/10.1145/2379057.2379081>
 - [26] Amy Edmondson. 1999. Psychological Safety and Learning Behavior in Work Teams. *Administrative Science Quarterly* 44, 2 (June 1999), 350–383. <https://doi.org/10.2307/2666999>
 - [27] Anne Edwards. 2005. Relational agency: Learning to be a resourceful practitioner. *International Journal of Educational Research* 43, 3 (Jan. 2005), 168–182. <https://doi.org/10.1016/j.ijer.2006.06.010>
 - [28] Jessica L. Feuston and Jed R. Brubaker. 2021. Putting Tools in Their Place: The Role of Time and Perspective in Human-AI Collaboration for Qualitative Analysis. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW2 (2021), 469:1–469:25. <https://doi.org/10.1145/3479856>
 - [29] Roger Fisher, William L. Ury, and Bruce Patton. 2011. *Getting to Yes: Negotiating Agreement Without Giving In*. Penguin. Google-Books-ID: W89fHCjZrcwC.
 - [30] Jay Folberg and Alison Taylor. 1991. *Mediation: A Comprehensive Guide to Resolving Conflicts Without Litigation*. Wiley. Google-Books-ID: Tn1Pz25XVGEC.
 - [31] Donelson R. Forsyth. 1990. *Group Dynamics*. Brooks/Cole Publishing Company. Google-Books-ID: VhNHAAAMAAMAJ.
 - [32] Yue Fu, Sami Foell, Xuhai Xu, and Alexis Hiniker. 2024. From Text to Self: Users' Perception of AIMC Tools on Interpersonal Communication and Self. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM, Honolulu HI USA, 1–17. <https://doi.org/10.1145/3613904.3641955>
 - [33] Eleni Georganta, Claudia Peus, and Jasmin Niess. 2024. Interactive technologies through the lens of team effectiveness: an interdisciplinary systematic literature review. *European Journal of Work and Organizational Psychology* 33, 2 (March 2024), 172–187. <https://doi.org/10.1080/1359432X.2023.2178904>
 - [34] Erving Goffman. 1959. *The presentation of self in everyday life*. Doubleday, Oxford, England.
 - [35] Yijie Guo, Ruhan Wang, Yuan-Ling Feng, Yao Lu, Xuezhu Wang, Fumihide Tanaka, and Haipeng Mi. 2024. ROMEO: Towards the Design of Robot with Haptic Mediation for Remote Conflict. In *Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems (CHI EA '24)*. Association for Computing Machinery, New York, NY, USA, 1–7. <https://doi.org/10.1145/3613905.3650781> 0 citations (Semantic Scholar/DOI) [2024-05-22] 0 citations (Crossref) [2024-05-22].
 - [36] Ariel Han, Xiaofei Zhou, Zhenyao Cai, Shenshen Han, Richard Ko, Seth Corrigan, and Kylie A. Peppler. 2024. Teachers, Parents, and Students' perspectives on Integrating Generative AI into Elementary Literacy Education. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM, Honolulu HI USA, 1–17. <https://doi.org/10.1145/3613904.3642438>
 - [37] Jeffrey T Hancock, Mor Naaman, and Karen Levy. 2020. AI-Mediated Communication: Definition, Research Agenda, and Ethical Considerations. *Journal of Computer-Mediated Communication* 25, 1 (March 2020), 89–100. <https://doi.org/10.1093/jcmc/zmz022>
 - [38] Steve Harrison and Paul Dourish. 1996. Re-place-ing space: the roles of place and space in collaborative systems. In *Proceedings of the 1996 ACM conference on Computer supported cooperative work (CSCW '96)*. Association for Computing Machinery, New York, NY, USA, 67–76. <https://doi.org/10.1145/240080.240193>
 - [39] Kevin Hogan, Annabelle Baer, and James Purtilo. 2021. Diplomat: A Conversational Agent Framework for Goal-Oriented Group Discussion. In *Contemporary Issues in Group Decision and Negotiation*, Danielle Costa Morais, Liping Fang, and Masahide Horita (Eds.). Springer International Publishing, Cham, 143–154. https://doi.org/10.1007/978-3-030-77208-6_11
 - [40] George Hope Chidziwisano, Alex Mariakakis, Susan Wyche, Vitumbiko Mafeni, and Esau Gideon Banda. 2021. NkhukuProbe: Using a Sensor-Based Technology Probe to Support Poultry Farming Activities in Malawi. In *Proceedings of the 4th ACM SIGCAS Conference on Computing and Sustainable Societies (COMPASS '21)*. Association for Computing Machinery, New York, NY, USA, 275–287. <https://doi.org/10.1145/3460112.3471962>
 - [41] Mo Houtti, Moyan Zhou, Loren Terveen, and Stevie Chancellor. 2025. Observe, Ask, Intervene: Designing AI Agents for More Inclusive Meetings. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. ACM, Yokohama Japan, 1–18. <https://doi.org/10.1145/3706598.3713838>
 - [42] Erzhen Hu, Md Aashikur Rahman Azim, and Seongkook Heo. 2022. Fluid-Meet: Enabling Frictionless Transitions Between In-Group, Between-Group, and Private Conversations During Virtual Breakout Meetings. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*. Association for Computing Machinery, New York, NY, USA, 1–17. <https://doi.org/10.1145/3491102.3517558>
 - [43] Hilary Hutchinson, Benjamin B. Bederson, Allison Druin, Catherine Plaisant, Wendy Mackay, Helen Evans, Heiko Hansen, Stéphane Conversy, Michel Beaudouin-Lafon, Nicolas Roussel, Loïc Lacomme, Björn Eiderbäck, Sinna Lindquist, Yngve Sundblad, and Bosse Westerglund. [n. d.]. Technology Probes: Inspiring Design for and with Families. ([n. d.]).
 - [44] Zainab Ifthikhar, Yumeng Ma, and Jeff Huang. 2023. "Together but not together": Evaluating Typing Indicators for Interaction-Rich Communication. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM, Hamburg Germany, 1–12. <https://doi.org/10.1145/3544548.3581248>
 - [45] Katherine Isbister, Hideyuki Nakanishi, Toru Ishida, and Cliff Nass. 2000. Helper agent: designing an assistant for human-human interaction in a virtual meeting space. In *Proceedings of the SIGCHI conference on Human Factors in Computing Systems*. ACM, The Hague The Netherlands, 57–64. <https://doi.org/10.1145/332040.332407>
 - [46] Daeun Jeong, Sungbok Shin, and Jongwook Jeong. 2025. Conversation Progress Guide : UI System for Enhancing Self-Efficacy in Conversational AI. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. ACM, Yokohama Japan, 1–11. <https://doi.org/10.1145/3706598.3714222>
 - [47] Heidi Jokinen. [n. d.]. Social Mediation - Working Towards Inclusion from Amidst Exclusion. ([n. d.]).
 - [48] Matthew Jörke, Yasaman S. Sefidgar, Talie Massachi, Jina Suh, and Gonzalo Ramos. 2023. Pearl: A Technology Probe for Machine-Assisted Reflection on Personal Data. In *Proceedings of the 28th International Conference on Intelligent User Interfaces (IUI '23)*. Association for Computing Machinery, New York, NY, USA, 902–918. <https://doi.org/10.1145/3581641.3584054>
 - [49] Fariba Karimi, Verónica C. Ramenzoni, and Petter Holme. 2014. Structural differences between open and direct communication in an online community. *Physica A: Statistical Mechanics and its Applications* 414 (Nov. 2014), 263–273. <https://doi.org/10.1016/j.physa.2014.07.037>
 - [50] Yael Keshet, Eran Ben-Arye, and Elad Schiff. 2013. The use of boundary objects to enhance interprofessional collaboration: integrating complementary medicine in a hospital setting. *Sociology of Health & Illness* 35, 5 (2013), 666–681. <https://doi.org/10.1111/j.1467-9566.2012.01520.x> eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1467-9566.2012.01520.x>.
 - [51] Vera Khovanskaya, Eric P.S. Baumer, Dan Cosley, Stephen Volda, and Geri Gay. 2013. "Everybody knows what you're doing": a critical design approach to personal informatics. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '13)*. Association for Computing Machinery, New York, NY, USA, 3403–3412. <https://doi.org/10.1145/2470654.2466467>
 - [52] Soomin Kim, Jinsu Eun, Changhoon Oh, Bongwon Suh, and Joonhwan Lee. 2020. Bot in the Bunch: Facilitating Group Chat Discussion by Improving Efficiency and Participation with a Chatbot. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3313831.3376785>
 - [53] Soomin Kim, Jinsu Eun, Joseph Seering, and Joonhwan Lee. 2021. Moderator Chatbot for Deliberative Discussion: Effects of Discussion Structure and Discussant Facilitation. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW1 (2021), 87:1–87:26. <https://doi.org/10.1145/3449161>
 - [54] Taenyun Kim, Maria D. Molina, Minjin (MJ) Rheu, Emily S. Zhan, and Wei Peng. 2023. One AI Does Not Fit All: A Cluster Analysis of the Laypeople's Perception of AI Roles. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA, 1–20. <https://doi.org/10.1145/3544548.3581340>
 - [55] Mateusz Tomasz Kot and Grzegorz Leszczyński. 2020. The concept of intelligent agent in business interactions: is virtual assistant an actor or a boundary object? *Journal of Business & Industrial Marketing* 35, 7 (Feb. 2020), 1155–1164. <https://doi.org/10.1108/JBIM-10-2018-0291>
 - [56] Kenneth Kressel. 2006. Mediation revisited. In *The handbook of conflict resolution: Theory and practice, 2nd ed.* Wiley Publishing, Hoboken, NJ, US, 726–756.
 - [57] Charlotte P. Lee. 2007. Boundary Negotiating Artifacts: Unbinding the Routine of Boundary Objects and Embracing Chaos in Collaborative Work. *Comput. Supported Coop. Work* 16, 3 (2007), 307–339. <https://doi.org/10.1007/s10606-007-9044-5>

- [58] Soohwan Lee, Seoyeong Hwang, and Kyungho Lee. 2025. Beyond Individual UX: Defining Group Experience(GX) as a New Paradigm for Group-centered AI. In *Companion Publication of the 2025 ACM Designing Interactive Systems Conference*. Association for Computing Machinery, New York, NY, USA, 357–362. <https://doi.org/10.1145/3715668.3736348>
- [59] Sung-Chul Lee, Jaeyoon Song, Eun-Young Ko, Seongho Park, Jihee Kim, and Juho Kim. 2020. SolutionChat: Real-time Moderator Support for Chat-based Structured Discussion. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3313831.3376609>
- [60] Chieh-Peng Lin, Hongwei He, Yehuda Baruch, and Blake E. Ashforth. 2017. The Effect of Team Affective Tone on Team Performance: The Roles of Team Identification and Team Cooperation. *Human Resource Management* 56, 6 (Nov. 2017), 931–952. <https://doi.org/10.1002/hrm.21810>
- [61] Dingdong Liu, Yujing Zhang, Bolin Zhao, Shuai Ma, Chuhan Shi, and Xiaojuan Ma. 2025. Scaffolded Turns and Logical Conversations: Designing Humanized LLM-Powered Conversational Agents for Hospital Admission Interviews. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. ACM, Yokohama Japan, 1–23. <https://doi.org/10.1145/3706598.3714196>
- [62] Xingyu Bruce Liu, Shitao Fang, Weiyan Shi, Chien-Sheng Wu, Takeo Igarashi, and Xiang 'Anthony' Chen. 2025. Proactive Conversational Agents with Inner Thoughts. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. ACM, Yokohama Japan, 1–19. <https://doi.org/10.1145/3706598.3713760>
- [63] Inês Lobo, Janin Koch, Jennifer Renoux, Inês Batina, and Rui Prada. 2024. When Should I Lead or Follow: Understanding Initiative Levels in Human-AI Collaborative Gameplay. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference (DIS '24)*. Association for Computing Machinery, New York, NY, USA, 2037–2056. <https://doi.org/10.1145/3643834.3661583>
- [64] Sus Lundgren and Dimitrios Gkouskos. 2013. Escaping the obvious: Skewing properties of interaction. *Nordes Conference Series* (June 2013). <https://dl.designresearchsociety.org/nordes/nordes2013/researchpapers/2>
- [65] Wayne G. Lutters and Mark S. Ackerman. 2002. Achieving safety: a field study of boundary objects in aircraft technical support. In *Proceedings of the 2002 ACM conference on Computer supported cooperative work (CSCW '02)*. Association for Computing Machinery, New York, NY, USA, 266–275. <https://doi.org/10.1145/587078.587116>
- [66] Karolina Mania. 2015. Online dispute resolution: The future of justice. *International Comparative Jurisprudence* 1, 1 (Nov. 2015), 76–86. <https://doi.org/10.1016/j.icj.2015.10.006>
- [67] Manqing Mao, Paishun Ting, Yijian Xiang, Mingyang Xu, Julia Chen, and Jianzhe Lin. 2024. Multi-User Chat Assistant (MUCA): a Framework Using LLMs to Facilitate Group Conversations. <https://doi.org/10.48550/arXiv.2401.04883> [cs].
- [68] Bernard S. Mayer. 2012. *The Dynamics of Conflict: A Guide to Engagement and Intervention*. John Wiley & Sons.
- [69] Jim McCambridge, John Witton, and Diana R. Elbourne. 2014. Systematic review of the Hawthorne effect: New concepts are needed to study research participation effects. *Journal of Clinical Epidemiology* 67, 3 (March 2014), 267–277. <https://doi.org/10.1016/j.jclinepi.2013.08.015>
- [70] Thomas Mildner, Gian-Luca Savino, Philip R. Doyle, Benjamin R. Cowan, and Rainer Malaka. 2023. About Engaging and Governing Strategies: A Thematic Analysis of Dark Patterns in Social Networking Services. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA, 1–15. <https://doi.org/10.1145/3544548.3580695>
- [71] Matthew B. Miles, A. Michael Huberman, and Johnny Saldana. 2014. *Qualitative Data Analysis*. SAGE. Google-Books-ID: 3CNrU6Csc.
- [72] Christopher W. Moore. 1996. *The Mediation Process: Practical Strategies for Resolving Conflict*. Wiley. Google-Books-ID: cRxHAAAAMAAJ.
- [73] Qianqian Mu, Marcel Borowski, Jens Emil Sloth Grønæk, Susanne Bødker, and Eve Hoggan. 2024. Whispering Through Walls: Towards Inclusive Backchannel Communication in Hybrid Meetings. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, 1–16. <https://doi.org/10.1145/3613904.3642419>
- [74] N. Nilsson. 1984. Shakey the Robot. <https://www.semanticscholar.org/paper/Shakey-the-Robot-Nilsson/476ba2a1c5204d46e420506afac4b0da6abb868>
- [75] Helen Nissenbaum. 2004. Privacy as Contextual Integrity. *Washington Law Review* 79, 1 (Feb. 2004), 119. <https://digitalcommons.law.uw.edu/wlr/vol79/iss1/10>
- [76] B. Oakley, R. Felder, R. Brent, and Imad H. Elhajj. 2004. Turning Student Groups into Effective Teams. https://www.semanticscholar.org/paper/Turning-Student-Groups-into-Effective-Teams-Oakley-Felder/93045b26a224b454f45f97a6af70746c79e2bbd1?utm_source=chatgpt.com
- [77] Jeesun Oh, Woosok Kim, Sungbae Kim, Hyeonjeong Im, and Sangsu Lee. 2024. Better to Ask Than Assume: Proactive Voice Assistants' Communication Strategies That Respect User Agency in a Smart Home Environment. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, 1–17. <https://doi.org/10.1145/3613904.3642193>
- [78] Omar Shaikh, Valentino Emil Chai, Michele Gelfand, Diyi Yang, and Michael S. Bernstein. 2024. Rehearsal: Simulating Conflict to Teach Conflict Resolution. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, 1–20. <https://doi.org/10.1145/3613904.3642159> 1 citations (Crossref) [2024-06-11].
- [79] Hyanghee Park, Daehwan Ahn, Kartik Hosanagar, and Joonhwan Lee. 2021. Human-AI Interaction in Human Resource Management: Understanding Why Employees Resist Algorithmic Evaluation at Workplaces and How to Mitigate Burdens. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. ACM, Yokohama Japan, 1–15. <https://doi.org/10.1145/3411764.3445304>
- [80] Hyanghee Park, Daehwan Ahn, and Joonhwan Lee. 2023. Towards a Metaverse Workspace: Opportunities, Challenges, and Design Implications. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM, Hamburg Germany, 1–20. <https://doi.org/10.1145/3544548.3581306>
- [81] Hyanghee Park, Daehwan Ahn, and Joonhwan Lee. 2024. Lessons From Working in the Metaverse: Challenges, Choices, and Implications from a Case Study. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM, Honolulu HI USA, 1–16. <https://doi.org/10.1145/3613904.3641972>
- [82] Sandra Petronio. 2002. *Boundaries of privacy: Dialectics of disclosure*. State University of New York Press, Albany, NY, US. Pages: xix, 268.
- [83] Vicente Peñarroja, Virginia Orengo, Ana Zornoza, Jesús Sánchez, and Pilar Ripoll. 2015. How team feedback and team trust influence information processing and learning in virtual teams: A moderated mediation model. *Computers in Human Behavior* 48 (July 2015), 9–16. <https://doi.org/10.1016/j.chb.2015.01.034>
- [84] Alexander Pohl, Vera Gehlen-Baum, and Francois Bry. 2011. Introducing Backstage—A Digital Backchannel for Large Class Lectures. *Interactive Technology and Smart Education* 8, 3 (2011), 186–200. <https://doi.org/10.1108/17415651111165410> Publisher: Emerald ERIC Number: EJ940665.
- [85] David Premack and Guy Woodruff. 1978. Does the chimpanzee have a theory of mind? *Behavioral and Brain Sciences* 1, 4 (Dec. 1978), 515–526. <https://doi.org/10.1017/S0140525X00076512>
- [86] Sara Pérez-Soler, Esther Guerra, and Juan de Lara. 2018. Collaborative Modeling and Group Decision Making Using Chatbots in Social Networks. *IEEE Software* 35, 6 (Nov. 2018), 48–54. <https://doi.org/10.1109/MS.2018.290101511> Conference Name: IEEE Software.
- [87] C. Ross, M. Terras, C. Warwick, and A. Welsh. 2011. Enabled backchannel: conference Twitter use by digital humanists. *Journal of Documentation* 67, 2 (March 2011), 214–237. <https://doi.org/10.1108/00220411111109449>
- [88] Kavous Salehzadeh Niksirat, Diana Korka, Hamza Harkous, Kevin Huguenin, and Mauro Cherubini. 2023. On the Potential of Mediation Chatbots for Mitigating Multiparty Privacy Conflicts - A Wizard-of-Oz Study. *Proceedings of the ACM on Human-Computer Interaction* 7, CSCW1 (2023), 142:1–142:33. <https://doi.org/10.1145/3579618> 4 citations (Semantic Scholar/DOI) [2024-05-22] 2 citations (Crossref) [2024-05-22].
- [89] Joseph Seering, Manas Khadka, Nava Haghighi, Tanya Yang, Zachary Xi, and Michael Bernstein. 2024. Chillbot: Content Moderation in the Backchannel. *Proc. ACM Hum.-Comput. Interact.* 8, CSCW2 (2024), 402:1–402:26. <https://doi.org/10.1145/3686941>
- [90] Joseph Seering, Robert Kraut, and Laura Dabbish. 2017. Shaping Pro and Anti-Social Behavior on Twitch Through Moderation and Example-Setting. In *Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing*. ACM, Portland Oregon USA, 111–125. <https://doi.org/10.1145/2998181.2998277>
- [91] Joseph Seering, Michal Luria, Geoff Kaufman, and Jessica Hammer. 2019. Beyond Dyadic Interactions: Considering Chatbots as Community Members. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, Glasgow Scotland UK, 1–13. <https://doi.org/10.1145/3290605.3300680>
- [92] Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askeel, Samuel R. Bowman, Newton Cheng, Esin Durmus, Zac Hatfield-Dodds, Scott R. Johnston, Shauna Kravec, Timothy Maxwell, Sam McCandlish, Kamal Ndousse, Oliver Rausch, Nicholas Schiefer, Da Yan, Miranda Zhang, and Ethan Perez. 2023. Towards Understanding Sycophancy in Language Models. <https://doi.org/10.48550/arXiv.2310.13548> arXiv:2310.13548 [cs].
- [93] Solace Shen, Petr Slovak, and Malte F. Jung. 2018. "Stop. I See a Conflict Happening.": A Robot Mediator for Young Children's Interpersonal Conflict Resolution. In *Proceedings of the 2018 ACM/IEEE International Conference on Human-Robot Interaction (HRI '18)*. Association for Computing Machinery, New York, NY, USA, 69–77. <https://doi.org/10.1145/3171221.3171248> 81 citations (Semantic Scholar/DOI) [2024-05-22] 50 citations (Crossref) [2024-05-22].
- [94] Donghoon Shin, Soomin Kim, Ruoxi Shang, Joonhwan Lee, and Gary Hsieh. 2023. IntroBot: Exploring the Use of Chatbot-assisted Familiarization in Online Collaborative Groups. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM, Hamburg Germany, 1–13. <https://doi.org/10.1145/3544548.3580930>

- [95] Donghoon Shin, Sangwon Yoon, Soomin Kim, and Joonhwan Lee. 2021. Blah-BlahBot: Facilitating Conversation between Strangers using a Chatbot with ML-infused Personalized Topic Suggestion. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems (CHI EA '21)*. Association for Computing Machinery, New York, NY, USA, 1–6. <https://doi.org/10.1145/3411763.3451771>
- [96] Ben Shneiderman. 2020. Human-Centered Artificial Intelligence: Reliable, Safe & Trustworthy. *International Journal of Human-Computer Interaction* 36, 6 (April 2020), 495–504. <https://doi.org/10.1080/10447318.2020.1741118> Publisher: Taylor & Francis _eprint: <https://doi.org/10.1080/10447318.2020.1741118>.
- [97] Georg Simmel. 1906. The Sociology of Secrecy and of Secret Societies. *Amer. J. Sociology* 11, 4 (Jan. 1906), 441–498. <https://doi.org/10.1086/211418> Publisher: The University of Chicago Press.
- [98] Jeff Sugarman, , and Jack Martin. 2011. Theorizing Relational Agency. *Journal of Constructivist Psychology* 24, 4 (Oct. 2011), 283–289. <https://doi.org/10.1080/10720537.2011.593455> Publisher: Routledge _eprint: <https://doi.org/10.1080/10720537.2011.593455>.
- [99] Xiaoqing Sun, Jingyi Wang, Yan Zhou, Suhan Wang, Yixuan Li, and Xipei Ren. 2024. CO-Coffee: A Technology Probe Study to Facilitate Coffee Breaks in Open Offices. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*. Association for Computing Machinery, New York, NY, USA, 1–7. <https://doi.org/10.1145/3613905.3651030>
- [100] Susan Leigh Star and James R. Griesemer. 1989. Institutional Ecology, 'Translations' and Boundary Objects: Amateurs and Professionals in Berkeley's Museum of Vertebrate Zoology, 1907–39. *Social Studies of Science* 19, 3 (1989), 387–420. <http://www.jstor.org/stable/285080>
- [101] Cheri Toledo and Sharon Peters. 2010. Educators' Perceptions of Uses, Constraints, and Successful Practices of Backchanneling. *in education* 16, 1 (2010). <https://doi.org/10.37119/ojs2010.v16i1.48>
- [102] Bruce W. Tuckman. 1965. Developmental sequence in small groups. *Psychological Bulletin* 63, 6 (1965), 384–399. <https://doi.org/10.1037/h0022100> Place: US Publisher: American Psychological Association.
- [103] Bruce W. Tuckman and Mary Ann C. Jensen. 1977. Stages of Small-Group Development Revisited. *Group & Organization Studies* 2, 4 (Dec. 1977), 419–427. <https://doi.org/10.1177/105960117700200404> Publisher: SAGE Publications.
- [104] Stephen Viller. 1991. The Group Facilitator: A CSCW Perspective. In *Proceedings of the Second European Conference on Computer-Supported Cooperative Work ECSCW '91*, Liam Bannon, Mike Robinson, and Kjeld Schmidt (Eds.). Springer Netherlands, Dordrecht, 81–95. https://doi.org/10.1007/978-94-011-3506-1_6
- [105] David W. Johnson and Roger T. Johnson. 2019. Cooperative Learning: The Foundation for Active Learning. In *Active Learning - Beyond the Future*, Silvio Manuel Brito (Ed.). IntechOpen. <https://doi.org/10.5772/intechopen.81086>
- [106] James A. Wall, John B. Stark, and Rheta L. Standifer. 2001. Mediation: A Current Review and Theory Development. *The Journal of conflict resolution* 45, 3 (2001), 370–391. <https://doi.org/10.1177/0022002701045003006> 203 citations (Crossref) [2024-05-29] Place: 2455 Teller Road, Thousand Oaks, CA 91320 Publisher: Sage Publications.
- [107] Dakuo Wang, Elizabeth Churchill, Pattie Maes, Xiangmin Fan, Ben Shneiderman, Yuanchun Shi, and Qianying Wang. 2020. From Human-Human Collaboration to Human-AI Collaboration: Designing AI Systems That Can Work Together with People. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems (CHI EA '20)*. Association for Computing Machinery, New York, NY, USA, 1–6. <https://doi.org/10.1145/3334480.3381069>
- [108] Qiaosi Wang. 2024. MUTUAL THEORY OF MIND FOR HUMAN-AI COMMUNICATION IN AI-MEDIATED SOCIAL INTERACTION. (2024).
- [109] Qiaosi Wang, Shan Jing, and Ashok K. Goel. 2022. Co-Designing AI Agents to Support Social Connectedness Among Online Learners: Functionalities, Social Characteristics, and Ethical Challenges. In *Designing Interactive Systems Conference*. ACM, Virtual Event Australia, 541–556. <https://doi.org/10.1145/3532106.3533534>
- [110] Ruotong Wang, Xinyi Zhou, Lin Qiu, Joseph Chee Chang, Jonathan Bragg, and Amy X. Zhang. 2025. Social-RAG: Retrieving from Group Interactions to Socially Ground AI Generation. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. ACM, Yokohama Japan, 1–25. <https://doi.org/10.1145/3706598.3713749>
- [111] Karl E. Weick. 1995. *Sensemaking in Organizations*. SAGE.
- [112] Thomas H. Weisswange, Hifza Javed, Manuel Dietrich, Malte F. Jung, and Nawid Jamali. 2024. Social Mediation through Robots – A Scoping Review on Improving Group Interactions through Directed Robot Action using an Extended Group Process Model. <https://doi.org/10.48550/arXiv.2409.06557> arXiv:2409.06557 [cs].
- [113] Harvey Whitehouse and Jonathan A. Lanman. 2014. The Ties That Bind Us: Ritual, Fusion, and Identification. *Current Anthropology* 55, 6 (Dec. 2014), 674–695. <https://doi.org/10.1086/678698>
- [114] Michael Williams and Tami Moser. 2019. The Art of Coding and Thematic Exploration in Qualitative Research. *International Management Review* (2019). <https://www.semanticscholar.org/paper/The-Art-of-Coding-and-Thematic-Exploration-in-Williams-Moser/c0a0c26ac41cb8beb337834e6c1e2f35b91d071d>
- [115] A Zander. 1979. The Psychology of Group Processes. *Annual Review of Psychology* 30, 1 (Jan. 1979), 417–451. <https://doi.org/10.1146/annurev.ps.30.020179.002221>
- [116] Chengbo Zheng, Yuheng Wu, Chuhan Shi, Shuai Ma, Jiehui Luo, and Xiaojuan Ma. 2023. Competent but Rigid: Identifying the Gap in Empowering AI to Participate Equally in Group Decision-Making. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA, 1–19. <https://doi.org/10.1145/3544548.3581131>

A Formative Study: Focus Group Protocol

Timeline (90 Minutes)

- **5 min - Part I: Overview** Introduction and Ground Rules
- **30 min - Part II: Group Interview** Self-Introductions and Teamwork Reflections
- **55 min - Part III: Co-Design & SeeSawBot Trial** Scenario-based interaction with prototype

Part I: Introduction (5 min)

Co-design an AI agent to improve team dynamics and personal growth through interactive scenarios.

Part II: Group Interview (30 min)

Prompts:

- Preferred name and academic background.
- Past experience with teamwork and mediation.
- Describe a memorable mediation experience.
- How did you involve third parties? What did you learn?

Part III: Role-Play Testing (55 min)

Interaction.

- *Channel Conversations*: Simulate group chats.
- *Direct Messages*: Private chat with SeeSawBot.

Scenarios.

- Scenario 1: First Meeting
- Scenario 2: First Deadline Approaching
- Scenario 3: Disagreement on Assignment
- Scenario 4: Final Deadline & Unequal Participation

Discussion Prompts. For each scenario, participants are asked to reflect on:

- Where did the AI respond (DMs vs. channel)?
- Who was addressed and was it appropriate?
- What was the function of each message (e.g., tension diffusion, conflict resolution)?
- Was the timing effective?

Personas.

- **Alex**: Organized, task-oriented, deadline-driven.
- **Taylor**: Creative, inconsistent, idea-driven.
- **Jordan**: Independent, technical, responsive in meetings.
- **Morgan**: Optimistic, overcommitted, last-minute deliverer.

Part IV: Wrap-Up (10 minutes)

- Did you message SeeSawBot privately?
- Under what circumstances would you use SeeSawBot?
- Any final thoughts on the session?

B Detailed Data Analysis - Formative Study

Our analysis approached frictions and divergences not as anomalies but as productive design signals, extending prior work in human-centered design and HCI [8, 9, 51]. We paid particular attention to moments where SeeSawBot's presence generated tension, such as mismatched expectations, conflicting preferences, or emergent discomfort. Two principles guided our analytic decisions: (a) divergences in participant accounts were treated as analytically significant as convergences, and (b) a friction was considered meaningful if it was expressed verbally, enacted behaviorally, or inferred from a mismatch between the two.

We analyzed two types of data: focus group transcripts and researcher memos, all transcribed or transformed into text. Following Miles et al.'s qualitative analysis framework [71] and Williams and Moser's thematic exploration framework [114], analysis proceeded as follows:

- (1) **Data Familiarization.** Two researchers (R1 and R2) independently read the full corpus of transcripts and memos multiple times to build fluency with the material. Analytic memos captured emergent observations about the chatbot's positioning in group, boundary-crossing across spaces, and support preferences.
- (2) **Coding for Frictions.** Researchers independently coded excerpts, flagging instances of (a) expressed tension (e.g., "*I want it to remind me, but not in public*"), (b) enacted mismatch (e.g., hesitation before responding to the bot), or (c) contradictions across participants (e.g., one requesting humor while another found it unprofessional). Raw codes included "called out in public" (embarrassment), "bot as bad guy" (outsourcing uncomfortable authority), and "tone mismatch" (empathetic vs. directive). Codes were documented in a versioned codebook with definitions and examples. We adopted a negotiated coding approach rather than calculating inter-rater reliability, emphasizing transparent documentation of interpretive differences. Disagreements were resolved through memoing and discussion. When necessary, R1 and R2 consulted the last author, who brings expertise in qualitative research and technology-mediated communication. This phase produced 152 initial codes, organized in a shared codebook.
- (3) **Theme Development.** Using the codebook, R1 and R2 met for six 60-minute sessions to discuss and refine themes, evaluating candidate themes for internal coherence, distinctiveness, and design relevance. R1 and R2 clustered related codes into broader categories. For example, codes such as "gentle reminder," "bot as bad guy," and "avoiding confrontation" were grouped under the theme *Authority Positioning Flexibility*, reflecting tensions around how assertive the AI should be. Similarly, "called out in public" and "duplication DM+channel" formed the theme *Boundary-Testing Provocations*, capturing discomfort with how the bot moved between public and private spaces. These themes did not propose solutions but identified recurring sites of tension that highlighted key design trade-offs. Researcher memos were triangulated with transcript data to ensure themes reflected both stated preferences and enacted behaviors.

- (4) **Traceability to Design Principles.** Finally, the first author (R1) met with the last author to connect themes to design principles. A traceability matrix linked raw excerpts to codes, themes, and principles, ensuring transparency in how frictional moments informed design implications. This process culminated in four design principles (DP1-DP4), each supported by clusters of codes and illustrative quotes.

This process ensured that the formative study extended beyond feasibility testing, producing analytically grounded insights that informed SeeSawBot's baseline features and iteration trajectory.

C Deployment Study: Post-Deployment Interview Questions

Experience with the Chatbot in Teamwork

- (1) Can you describe a time when the chatbot reminded or helped coordinate your team about a task, role, or deadline? What did you do?
- (2) What was your first reaction to the bot's message?
- (3) Did you or your teammates respond? Why or why not?
- (4) Did it feel like the bot understood the timing or urgency of that moment?
- (5) Would it have been more effective in DM or channel?
- (6) Was there a time when the chatbot got someone's task or role wrong? How did your team handle it?
 - How did that person react?
 - Did anyone correct the bot? Or ignore it?
 - Did this affect trust in the bot going forward?
- (7) If someone was quiet or struggling, did the chatbot's message change how your team responded to them?
 - Do you remember what the bot said?
 - Did the message make you see the person differently?
 - Did it prompt any actions or conversations from your team?

Perceptions of Teammates and Social Dynamics

- (1) Was there a time the bot highlighted someone's action or inaction? How did your team respond?
- (2) Did it create any awkwardness or discussion afterward?
- (3) Would you prefer this kind of message in DM or public?
- (4) Did that moment change how you saw that teammate or others in the group?
 - Can you give a before/after example of your perception shift?
 - Did it make you more sympathetic? More critical? Why?
- (5) More generally, did the chatbot ever shape how you interpreted your teammates' roles, emotions, or contributions?
- (6) Do you think the chatbot understood your team's interaction patterns (team vibe), like who was active, who was quiet, or how people got along?
 - Were there moments it seemed aware of the social dynamics?
 - Did it ever misread the vibe? In what way?
 - If it got it right, how could you tell?
- (7) Did you ever feel like the chatbot was taking someone's side, maybe yours, or someone else's, during a disagreement?

- What made you feel that way?
 - Did it seem intentional, or a side effect of its behavior?
 - How did others respond?
- (8) How did the bot's presence shape how comfortable people felt expressing disagreement or offering critique?
 - Did anyone seem more cautious or more bold?
 - Did the bot ever say something during tension? Was that helpful or awkward?
 - (9) Looking across your experience, do you think the chatbot helped your team stay on the same page, or did it sometimes cause confusion or tension?
 - Can you give an example of each?
 - Did its presence make team alignment easier or harder?

Communication, Behavior, and Interpretation

- (1) How did you decide whether to talk to the chatbot in a DM or in the team channel? What influenced that choice?
- (2) Was that a strategic decision or just instinctive?
- (3) Did you worry about what others might think if you talked to the bot?
- (4) Can you describe a time when you changed how you communicated in Slack because you were aware the chatbot might respond or learn?
 - Did you avoid saying something? Or add extra explanation?
 - Was that a one-time thing or recurring?
- (5) In what ways, if any, did teammates act differently knowing the chatbot could message them privately?
 - Did people seem more careful or more responsive in Slack?
 - Were there any reactions like surprise or frustration?
- (6) Did you ever try to teach, redirect, or fine-tune the chatbot? For example, by helping it better understand you, your role, or how your team works?
 - What made you do that? What did you hope it would learn?
 - Did you feel it worked or not?
- (7) Was there a moment when the bot brought something from a DM into the team channel (or vice versa)?
 - How did you feel about that? Did it cross a boundary?
 - Did you wish you had more control over that behavior?
- (8) Looking back, how would you describe your dynamic with the chatbot?
 - Like a tool? A teammate? A distraction?
 - Did that change over time?
- (9) How did its presence shape your team's vibe?
 - If your team had a "vibe"? Was it focused? chaotic? chill? tense?
 - Did the chatbot fit into that vibe, or clash with it?

Future Imagination and Design

- (1) In an ideal world, what role should an ideal AI (more intelligent) play in a group like yours? What is missing in the current bot?
- (2) What social awareness and contextual understanding would make a chatbot more in sync with your team vibe (team's dynamics or interaction pattern)?

- (3) Should the chatbot intervene in social situations like tension, silence, or conflict? Why or why not?
- (4) If a more intelligent AI existed, in what kinds of situations would you want to invite it into your group chat?
- (5) If you wouldn't invite it, what would hold you back?

D Detailed Data Analysis - Deployment Study

We analyzed data from the 8-week deployment, comprising four rounds of bi-weekly surveys and nine post-deployment interviews, using an research-question-driven thematic analysis [71]. Following established practice in HCI (e.g., [36, 44, 63, 70]), we adopted a top-down approach in which each RQ (information flows, relational roles, temporal dynamics) served as an analytic lens. Combining surveys and interviews allowed us to capture breadth through longitudinal snapshots and depth through retrospective accounts, triangulating across data types to trace evolving perceptions of cross-space mediation. Our analytic process unfolded in four stages:

- (1) **Data Familiarization.** All four researchers reviewed survey responses (n=223 across four waves) and interview transcripts, making independent notes. A shared memoing document captured early impressions of trust, authority, and group processes.
- (2) **Initial Coding.** To distribute workload, two researchers focused primarily on survey data and three on interview data, each applying inductive descriptive codes. For interviews, each transcript was independently coded by two researchers, followed by six 90-minute comparison sessions in which coders discussed discrepancies and refined codes until reaching agreement. Example interview codes included "safe space in DM," "called out by bot," "neutral teammate," and "spammy interventions." Survey data were likewise double-coded, with twelve 120-minute meetings used to compare interpretations and consolidate codes. Example survey codes included "bot boosted efficiency," "flagged interaction patterns," and "direct discussion back on task." In addition, the full team held four 60-minute cross-check sessions to review overlapping subsets, align coding practices, and calibrate the evolving codebook. During these sessions, related codes were modified or merged. For instance, "absorbing blame" and "taking pressure off" were combined into a single consolidated code *bot absorbs pressure*. Together, surveys provided distributed snapshots across teams while interviews offered interpretive depth. Coding them in tandem supported cross-source comparison. This phase produced 253 distinct codes.
- (3) **Theme Development.** Researchers met jointly to cluster codes into higher-level categories within each RQ. For example, "delegating reminders," "bot absorbs pressure," and "avoiding being misunderstood" converged into the theme *AI as Social Buffer*. Disagreements were resolved through iterative discussion, with themes revised until consensus was reached. To ensure rigor, all candidate sub-themes were iteratively reviewed in six joint sessions (90-120 minutes each). During these meetings, researchers compared how codes clustered, challenged overlaps, and refined naming until consensus was reached. We checked that themes were

internally coherent, distinct from one another, and grounded in multiple data sources (survey and interview).

- (4) **Thematic Refinement and Validation.** The first author met weekly with the last author, an experienced qualitative researcher, to iteratively refine theme boundaries and ensure analytic coherence. Themes were checked for internal consistency, distinctiveness, and grounding in both survey and interview data. Cross-case comparisons (between teams

and across time) were used to examine how themes shifted across Tuckman's group development stages [102, 103]. The final thematic schema traced how participants negotiated autonomy, agency, and emotional labor across DM-channel boundaries.

This staged, RQ-driven process ensured transparency and rigor in how raw survey and interview data were synthesized into the findings reported in **Section 7**.